Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Alessandro Gori*

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Abstract

[To be continued. . .]

- Appendix on s-wave SC ghost simulations on HM
- Set ComputerModern font in all CairoMakie figures

Contents

*a.gori23@studenti.unipi.it / nepero27178 @ GitHub

List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued…]

Chapter 1

The normal phase

In this chapter we apply the HF methods sketched in Sec. **??** to the "normal phase". We are studying the model in a state were no symmetry of those enumerated in Tab. **??** is broken. Even though this phase may appear not so interesting, it actually shows a significant physical difference with respect to the conventional perfectly degenerate Fermi gas.

HF methods applied in this context will lead us to a self-consistent hopping renormalisation effect. For certain values of the non-local attraction V and the doping level, a complete Mott localisation effect takes place. The resulting bands shrinking and flattening is the main net effect of the presence of $\hat{H}_V$ in the EHM.

This chapter is divided in three main parts. First, we discuss the symmetries of the normal phase, following the general discussion of Chap. **??**, and apply HF methods to derive the self-consistent approximated hamiltonian. Second, we impose some theoretical constraints on the self-consistent solution. Lastly, we present numeric results.

1.1 The "normal" phase and its symmetries

In the normal phase no symmetry is broken. Thus, the HF wavefunction approximating the ground state must exhibit the full $\mathrm{U}(2) \otimes \mathrm{C}_4 \otimes \mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$ symmetry discussed in Sec. **??**. We take a slight detour from this, and instead allow for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\rightarrow} \mathrm{C}_2$$

This is the only potential SSB we consider. It will turn out in Sec. 1.5.2 that even allowing for breaking of rotational symmetry, the ground-state actually does not break C_4. This can be seen as a convolute way to confirm that there is not a normal-nematic phase. While this may seem as an useless complication, the derivation will set grounds for actual nematic effects found in the superconducting phase.

Our HF method relies on the application of Wick's theorem on the quartic part of the interacting hamiltonian, expressed in Eq. (**??**). We will decompose the quartic operators according to Eqns. (2.2), (**??**). Recalling Tabs. **??** and **??**, for the normal phase we must consider the following contributes to the EVs of Eqns. (**??**), (**??**), (**??**):

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

As we will see in detail and is summarised in Tab. 1.1, Hartree contractions give a chemical potential shift, while the unique Fock contraction effectively renormalises the bare bands.

In the normal phase, spin is a pure physical degeneracy, being the $\uparrow$ and $\downarrow$ sectors separated; moreover, $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. We will perform all computations leaving σ explicit and using this consideration at the end.

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalization and resymmetrization

Table 1.1 | Relevant Wick's contractions and net effect in the normal phase.

1.2 Hartree effects: chemical potential renormalization

Let us start by the analysis of the Hartree channel. The Hartree channel is defined as operator resulting from summing Hartree-like terms originating from Wick's decomposition of the quartic parts of $\hat{H}$. Each quartic term decomposes as a sum of fully contracted pairs, as in Eq. (**??**).

In the final hamiltonian, the net effect originating from Hartree terms will be a renormalisation of μ. To derive analytically this effect is only important to the ends of computing correctly the free energy density of the approximated ground state. As is discussed in Sec. **??**, in the context of numerical simulations at fixed density, the chemical potential is determined self-consistently at each step; thus any net effect that shifts μ is effectively ignored within (and limitedly to) the iterative algorithm.

1.2.1 Hartree channel in the normal phase: analytic derivation

The quartic part of $\hat{H}$ is given by the the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. Applying HF methods to them, we get the decomposition of Eqns. (2.2), (**??**). Let us deal with the Hartree sector of both separately.

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in Hartree channel as specified in Eq. (2.2)

$$\hat{H}_U \simeq U \sum_i \left[\langle \hat{n}_{i\uparrow}\rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow}\langle \hat{n}_{i\downarrow}\rangle - \langle \hat{n}_{i\downarrow}\rangle\langle \hat{n}_{i\uparrow}\rangle\right]$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma}\rangle = n \tag{1.1}$$

which gives

$$\begin{aligned}\hat{H}_U &\simeq nU \sum_i \left[\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}\right] - U \sum_i n^2 \\ &= nU \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/U}\end{aligned} \tag{1.2}$$

being $\hat{N}$ the total number operator and $E^{(\mathrm{N})}_{\mathrm{H}/U}$ the "contraction energy" (also known as double counting term). In Sec. 1.2.2 we deal with it. The chemical potential is corrected by

$$\mu \to \mu - nU \tag{1.3}$$

The non-local attraction further corrects μ.

Non-local attraction. The non-local attraction is divided in two sectors, as in Eq. (??). From Eq. (??), its Hartree-like terms are:

$$\hat{H}_V \simeq \overbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\sigma}+\hat{n}_{i\sigma}\langle\hat{n}_{j\sigma}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle\right]}^{\text{s.s.}} \\ \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma}\langle\hat{n}_{j\bar{\sigma}}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right]}_{\text{o.s.}}+(\text{all the rest}) \tag{1.4}$$

where "all the rest" collects all non-Hartree terms. Recalling the Ansatz of Eq. (1.1), we get

$$\hat{H}_V \simeq \underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{n}_{j\sigma}+\hat{n}_{i\sigma}\right]}_{\text{s.s.}}\underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma}\right]}_{\text{o.s.}}-E^{(\mathrm{N})}_{\mathrm{H}/V}+(\text{all the rest})$$

with $E^{(\mathrm{N})}_{\mathrm{H}/V}$ the normal state shift to energy brought by $\hat{H}_V$. Both sums above give a number operator. Note that we sum over NNs: each site is connected to $z = 4$ sites. Including this combinatorial factor, we have

$$\hat{H}_V \simeq -2nzV\times\hat{N}-E^{(\mathrm{N})}_{\mathrm{H}/V}+(\text{all the rest}) \tag{1.5}$$

This accounts for the final Hartree shift of μ,

$$\mu\to\mu+2znV \tag{1.6}$$

Considering the net shift to μ due to both interactions by combining Eqns. (1.3), (1.6), we get the final RMP anticipated in Tab. 1.1,

$$\tilde{\mu}\equiv\mu+n(2zV-U) \tag{1.7}$$

This result will remain valid throughout all phases discussed in this text. For the AF phase, Sec. ??, the SDW character leaves this shift untouched. The SC phase, Sec. 2.4.1, we will discuss is inherently translational invariant, thus the computation will be the same as here.

1.2.2 Hartree contraction energy

The double counting terms accounting for energy shifts in the Hartree channel are, as anticipated:

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/U} &= \sum_i\langle\hat{n}_{i\downarrow}\rangle\langle\hat{n}_{i\uparrow}\rangle \\ &= L_xL_y\times n^2U \end{aligned} \tag{1.8}$$

for the local repulsion and

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/V} &= -V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle+\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right] \\ &= -2V\sum_{\langle ij\rangle}\sum_{\sigma}n^2 \\ &= -L_xL_y\times 2zn^2V \end{aligned} \tag{1.9}$$

for the non-local attraction. In the last passage, we used Eq. (??) and spin degeneracy. The contraction energy is an extensive quantity. When computing total free energy density, we will use the contraction energy per site.

1.3 Fock effects: bare bands renormalization

Analysis of Fock terms will lead us to the effect that makes the normal phase interesting, when compared to the free Fermi gas on the square lattice. From Wick's decomposition of $\hat{H}_V$, the unique allowed Fock term comes from the same-spin sector. This selection rules is implied by the SU(2) symmetry of the normal phase. Then, from Eq. (**??**):

$$\hat{H}_V \simeq (\text{Hartree channel}) + V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \tag{1.10}$$

Note the $+$ sign in front of the displayed term: it originates from the $-$ sign in Fock term of Eq. (**??**). A bond-wise hopping amplitude can be defined,

$$\tilde{t}_{ij\sigma} \equiv t - V \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle$$

The effective kinetic hamiltonian is given by

$$\begin{aligned} \hat{H}_{\tilde{t}} &= \hat{H}_t + V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \text{h.c.} \right] \\ &= - \sum_{\langle ij \rangle} \sum_{\sigma} \left[\tilde{t}_{ij\sigma} \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] \end{aligned}$$

To proceed, it is useful to move to reciprocal space.

1.3.1 Fock channel in the normal phase: analytic derivation

In reciprocal space, the effective hopping must be transformed as well. Consider the Fourier Transform $\mathcal{F}$ given in Eq. (**??**), applied to the displayed part of Eq. (1.10),

$$\begin{aligned} &V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ &\quad \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{1.11}$$

Here, the 2 prefactor comes from recognizing that the first two operators in square brackets in the first line generate identical contributions to the full sum (as is easy to see by performing trivial variables changes); the double counting energy shift is given by

$$E^{(\mathrm{N})}_{\mathrm{F}/V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \tag{1.12}$$

Up to this point, this derivation is general and will be re-used in next chapters. Here the specificity of the physical phase settles in: if we assume the normal state to be a free degenerate Fermi gas with renormalized bands $\tilde{\epsilon}_{\mathbf{k}}$, we get

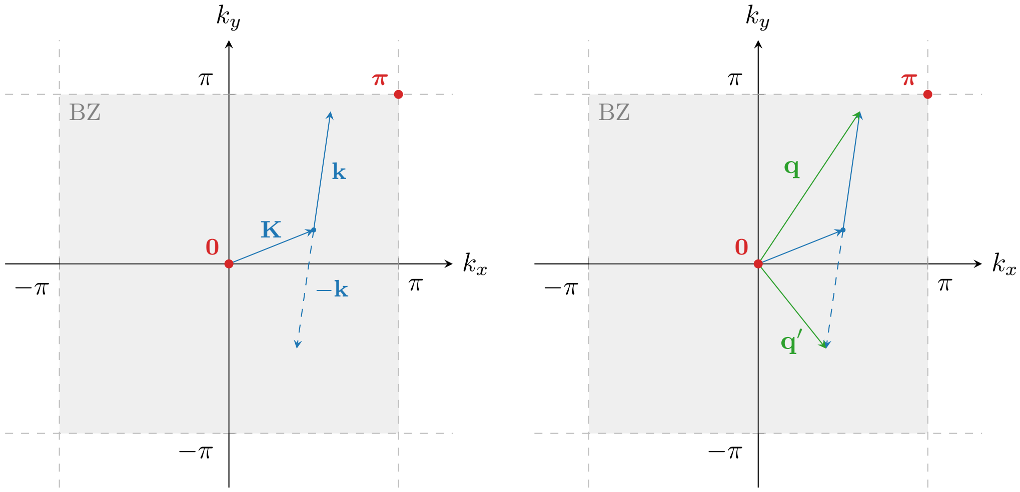
$$\langle \hat{c}^\dagger_{\mathbf{k}_1\sigma_1} \hat{c}_{\mathbf{k}_2\sigma_2} \rangle = \delta_{\mathbf{k}_1=\mathbf{k}_2} \delta_{\sigma_1=\sigma_2} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \tag{1.13}$$

being f the Fermi-Dirac distribution,

$$f(\epsilon; \beta, \mu) \equiv \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

It follows that Eq. (1.11) becomes

$$\begin{aligned} &V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right] \\ &\quad \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos(2k_x) + \cos(2k_y) \right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{1.14}$$



a | Generic vectors.   b | Change of variables.

Figure 1.1 | Representation of the variable change employed to pass from Eq. (1.14) to (1.15). In Fig. 1.1a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in$ BZ. In Fig. 1.1b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

since the only contribution comes from $\mathbf{k}' = -\mathbf{k}$.

Let now $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. A representation of this variables change is given in Fig. 1.1. Being for $\ell = x, y$

$$\begin{aligned} \delta q_\ell &\equiv q_\ell - q'_\ell \\ &= (K_\ell + k_\ell) - (K_\ell - k_\ell) \\ &= 2k_\ell \end{aligned}$$

we reduce Eq. (1.11) to

$$V \sum_{\langle ij \rangle} \sum_\sigma \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right]$$

$$\overset{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma \left[\cos(\delta q_x) + \cos(\delta q_y) \right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.15}$$

Recall now the result of the symmetry channels decomposition of Eq. (??),

$$\cos(\delta q_x) + \cos(\delta q_y) = \frac{1}{2} \sum_\gamma \varphi^{(\gamma)}_{\mathbf{q}} \varphi^{(\gamma)*}_{\mathbf{q}'} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$$

Thanks to this handy property, we reduce Eq. (1.15) to

$$V \sum_{\langle ij \rangle} \sum_\sigma \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right]$$

$$\overset{\mathcal{F}}{=} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right] \times V \sum_{\mathbf{q}' \in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.16}$$

where we defined the EV:

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \qquad u_{\mathbf{q}\sigma} = u^*_{\mathbf{q}\sigma} \tag{1.17}$$

The numeric value of $u_{\mathbf{q}\sigma}$ is given by the Fermi-Dirac distribution, but leaving it implicit we can re-use this computation later. Let $u^{(\gamma)}_\sigma$ be its γ-wave component,

$$u^{(\gamma)}_\sigma = \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. ??
$d_{x^2-y^2}$-wave	$u^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. ??

Table 1.2 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

The bare bands $\epsilon_{\mathbf{k}}$ are s^*-wave symmetric,

$$\epsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y) = -4t\varphi_{\mathbf{k}}^{(s^*)}$$

We used one function of those listed in Tab. ??. The renormalised bands $\tilde{\epsilon}_{\mathbf{k}}$ can exhibit a more complex symmetry. In the normal phase inversion symmetry is unbroken, thus

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

It follows that antisymmetric harmonics are zero,

$$u_\sigma^{(p_\ell)} = 0 \quad \text{for} \quad \ell \in \{x, y\}$$

The remaining s^* and d symmetries give legitimate HFPs with the respective self-consistency equations, reported in Tab. 1.2.Spin index is omitted due to spin degeneracy.

The two HFPs $u_\sigma^{(s^*)}$ and $u_\sigma^{(d)}$ are in general non-zero, since we are allowing for $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$. Note that when imposing C_4 invariance the $u_\sigma^{(d)}$ HFP vanishes, while in general $u_\sigma^{(s^*)}$ is non-zero. From these considerations Eq. (1.16) becomes

$$V \sum_{\langle ij \rangle} \sum_\sigma \left[\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle \right]$$
$$\overset{\mathcal{F}}{=} V \sum_{\mathbf{q}\in\mathrm{BZ}} \sum_\sigma \left[u_\sigma^{(s^*)} \varphi_{\mathbf{q}}^{(s^*)*} + u_\sigma^{(d)} \varphi_{\mathbf{q}}^{(d)*} \right] \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} - E_{\mathrm{F}/V}^{(\mathrm{N})} \tag{1.18}$$

Let

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V(\cos k_x + \cos k_y) + u_\sigma^{(d)} V(\cos k_x - \cos k_y) \tag{1.19}$$

This equation defines the bands renormalisation anticipated in Tab. 1.1, and is actually spin-independent. The fact that we have both s^* and a d-wave components introduces an anisotropy that is discussed in next section. However, we anticipate here that, as will result from Sec. 1.5.2, the d-wave harmonic will be suppressed. For now we keep this component ignoring these anticipations.

1.3.2 Fock contraction energy

Finally, let us derive the Fock contraction energy by expanding Eq. (1.12):

$$\begin{aligned} E_{\mathrm{F}/V}^{(\mathrm{N})} &= \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma [\cos(\delta q_x) + \cos(\delta q_y)] \langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} \rangle \langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} \rangle \\ &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^* \right] \\ &= L_x L_y \times \frac{V}{2} \sum_\sigma \left[\left| u_\sigma^{(s^*)} \right|^2 + \left| u_\sigma^{(d)} \right|^2 \right] \end{aligned} \tag{1.20}$$

This expression gives the Fock contraction energy coming from the $\hat{H}_V$ operator. The squared spectral amplitude of $u_{\mathbf{k}\sigma}$ here is summed, and contributes to contraction energy linearly in V. This suggests that a mixed-symmetry band could be energetically preferred.

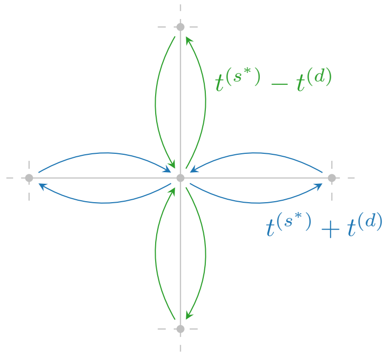


Figure 1.2 Graphic rendition of anisotropic hopping, with a s^*-wave component (symmetric by $\pi/$ rotations) and a d-wave component (antisymmetric).

1.3.3 Nematicity: extending hopping to d-wave bands

For the normal phase, there is no reason to expect nematic effects, nor to exclude them. They are observed in superconducting phase [23]. Thus, we allow the possibility.

From Tabs. **??** and **??** we can inverse-Fourier transform the form factors to get the full effective hopping,

$$\tilde{t}_{ij\sigma} = \sum_{\gamma \in \{s^*, d\}} t_\sigma^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

with

$$t_\sigma^{(s^*)} = t - \frac{u_\sigma^{(s^*)} V}{2} \qquad t_\sigma^{(d)} = -\frac{u_\sigma^{(d)} V}{2}$$

This is what we intend for "nematic effects". The possibility that planar rotational symmetry is broken in such a way that the kinetic operators in x and y directions differ. A schematics of anisotropic hopping is given in Fig. 1.2: for each bond linking two sites, the hopping parameter can be a positive or negative combination of the two components, whether the link direction is vertical or horizontal. The solution needs to be invariant under exchange $x \leftrightarrow y$, thus the actual SSB is $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$.

The fact that hopping is here anisotropic does not break full translational invariance $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$. Imposing a free degenerate Fermi gas ground state in Eq. (1.13) leads to a uniform density. This is proven by Fourier-transforming the site number operator according to the convention of Eq. (**??**),

$$\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma}$$

Then the on-site density EV is

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

A perfectly degenerate Fermi gas is always translational invariant and thus cannot exhibit non-uniform symmetry. The fact that hopping is allowed to be anisotropic does not interfere with that.

Note that for null HFPs,

$$u_\sigma^{(s^*)} = 0 \qquad u_\sigma^{(d)} = 0$$

we have, recalling Tab. **??**:

$$\tilde{t}_{\mathbf{r}\mathbf{r}'\sigma} = \frac{t}{2} \left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right]$$

The fact that $t/2$ appears here instead of t is a consequence of the equivalence:

$$-t \sum_{\langle ij \rangle} \sum_\sigma \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \mathrm{h.c.} \right] = -\frac{t}{2} \sum_{i,j \in \mathcal{S}} \sum_\sigma \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \mathrm{h.c.} \right]$$

where we including a prefactor that corrects double-counting. The RMP $\tilde{t}_{ij\sigma}$ must be interpreted like the right-hand side of the above equation, with the kinetic operator becoming:

$$\hat{H}_{\tilde{t}} \equiv -\frac{1}{2}\sum_{i,j\in\mathcal{S}}\sum_{\sigma}\left[\tilde{t}_{ij\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + \text{h.c.}\right]$$

To limit the sum to NNs is a matter of the symmetry structure of the function $\tilde{t}_{ij\sigma}$.

Recall the functions of Tab. **??**. We have the identities:

$$\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} = \varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} + \varphi^{(d)}_{\mathbf{r}\mathbf{r}'}$$
$$\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} = \varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} - \varphi^{(d)}_{\mathbf{r}\mathbf{r}'}$$

Then we can define the renormalised hopping on each direction as:

$$\tilde{t}^{(x)} \equiv t - \frac{u^{(s^*)} + u^{(d)}}{2}V$$
$$\tilde{t}^{(y)} \equiv t - \frac{u^{(s^*)} - u^{(d)}}{2}V$$

Both components depend linearly on V. The HFP $u^{(s^*)}$ is a measure of the direction-averaged hopping shift with respect to V, while $u^{(d)}$ measures the anisotropy,

$$u^{(s^*)} = \frac{\left(t-\tilde{t}^{(x)}\right) + \left(t-\tilde{t}^{(y)}\right)}{V} \tag{1.21}$$
$$u^{(d)} = \frac{\tilde{t}^{(x)} - \tilde{t}^{(y)}}{V} \tag{1.22}$$

It follows that if $u^{(d)} > 0$, the hopping in direction x is boosted while in direction y is damped. If $u^{(d)} < 0$, the contrary.

This discussion is general, and not specific to the normal phase. In next chapters we will interpret phase renormalisation as in this section.

1.4 Normal free energy density

In the end, we want to compare phases. Thus we need a way to extract the free energy density of the approximated ground-state for each phase. The procedure sketched here will be re-used in next chapters. Let

$$f^{(\mathrm{N})} \equiv \frac{F^{(\mathrm{N})}_{\mathrm{MFT}}}{L_x L_y}$$

being F_{MFT} the total free energy. Consider the complete hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k},\sigma}\tilde{\epsilon}_{\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} + E^{(\mathrm{N})}_{\mathrm{c}} - \tilde{\mu}\hat{N}$$

where $E^{(\mathrm{N})}_{\mathrm{c}}$ is the (negative) total contraction energy for the normal phase,

$$E^{(\mathrm{N})}_{\mathrm{c}} \equiv -\left[E^{(\mathrm{N})}_{\mathrm{H}/U} + E^{(\mathrm{N})}_{\mathrm{H}/V} + E^{(\mathrm{N})}_{\mathrm{F}/V}\right]$$

having included the various contraction energies from Eqns. (1.8), (1.9) and (1.20).

Let us now take together all terms. From now on, we will omit spin index σ, making use of the always present $\mathbb{Z}_2$ spin inversion symmetry discussed in Sec. 1.1. For a free Fermi gas, the free energy density is made of four contributions:

1. The free particles ground state energy density,

$$e^{(\mathrm{N})}_{\mathrm{g}} = \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

2. The contraction energy density,

$$\begin{aligned} e_{\mathrm{c}}^{(\mathrm{N})} &\equiv \frac{E_{\mathrm{c}}^{(\mathrm{N})}}{L_x L_y} \\ &= -n^2 U + V\left[2zn^2 - \left|u^{(s^*)}\right|^2 - \left|u^{(d)}\right|^2\right] \end{aligned}$$

3. The entropic contribution to free energy

$$e_s^{(\mathrm{N})} \equiv -Ts$$

with $T = 1/k_B\beta$ and s the free Fermi gas entropy density for a single spin sector [29],

$$s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}))\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right] \tag{1.23}$$

We use the identity:

$$\ln\left(1 - \frac{1}{e^x + 1}\right) = x + \ln\left(\frac{1}{e^x + 1}\right)$$

and, substituting $x = \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})$, reduce the above expression to:

$$\begin{aligned} s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[&\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \\ &- \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \Big] \end{aligned}$$

Second and fourth terms cancel out, and multiplying by $-2T$ (taking care of spin degeneracy) we have

$$e_s^{(\mathrm{N})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. The chemical potential correction

$$e_n^{(\mathrm{N})} = \mu n$$

Note here that we are using the "physical" value of μ, not its RMP counterpart $\tilde{\mu}$. The reason for this is discussed in Sec. ??.

Thus, composing everything

$$\begin{aligned} f^{(\mathrm{N})} &\equiv e_{\mathrm{g}}^{(\mathrm{N})} + e_{\mathrm{c}}^{(\mathrm{N})} + e_s^{(\mathrm{N})} + e_n^{(\mathrm{N})} \\ &= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \left[\mu + n(2zV - U)\right] n \end{aligned}$$

In Sec. ?? we argued that our iterative scheme computes $\tilde{\mu}$, not μ. Using Eq. (1.7), we see

$$\mu + n(2zV - U) = \tilde{\mu}$$

Then, the final expression is:

$$f^{(\mathrm{N})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \tilde{\mu} n \tag{1.24}$$

including all effects.

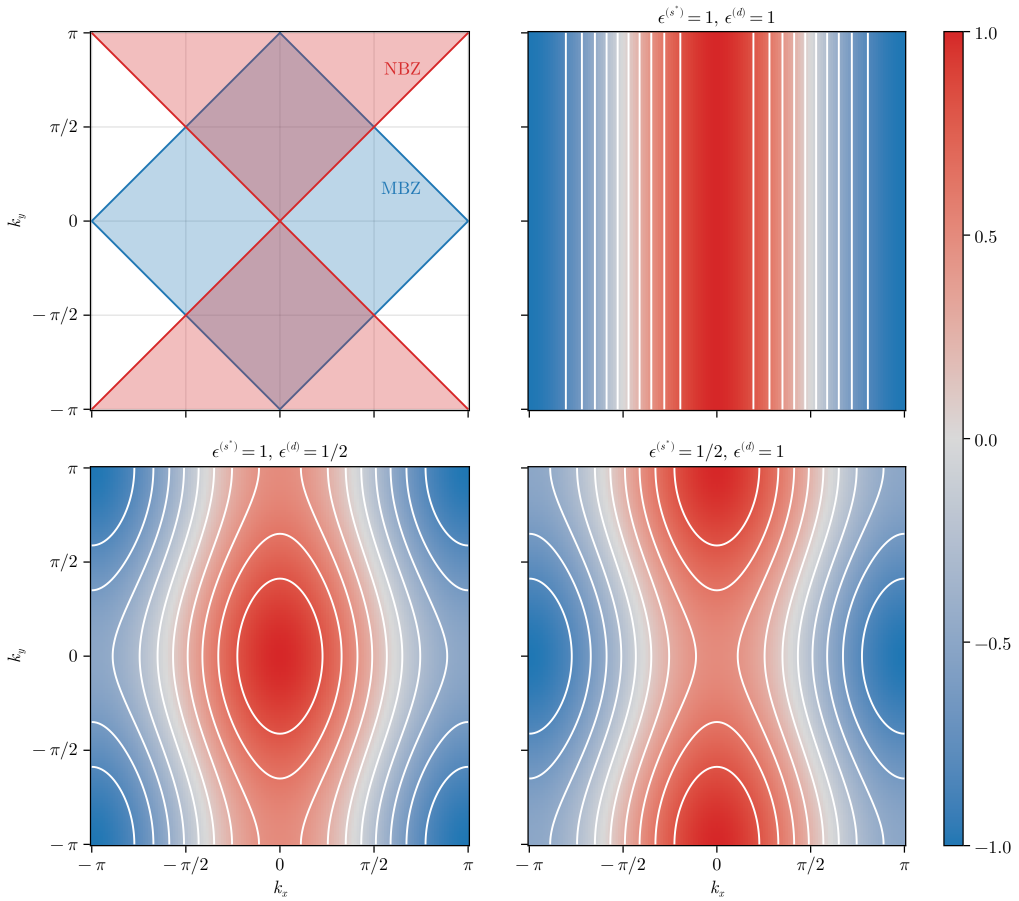


Figure 1.3 Theoretical representation of mixed-symmetry nematic bands for the normal phase. The white contour represent level lines. In top-left panel, the MBZ and NBZ are depicted. While these plots only account for positive-defined components, in text is described how to map each one on its counterpart with different sign.

1.5 Theoretical constraints on the normal phase HFPs

[Move this section to appendix?]

Within this section we discuss some theoretical constraints on the two HFPs of the normal phase. The discussion is divided in two parts: first we consider the simpler case of unviolated C_4 symmetry, then we move to anisotropic hopping and $C_4 \overset{\text{SSB}}{\to} C_2$.

Before starting, it is useful to spend a few words on the spatial structure of a mixed $s^* \otimes d$-wave band. Let in general for notational simplicity

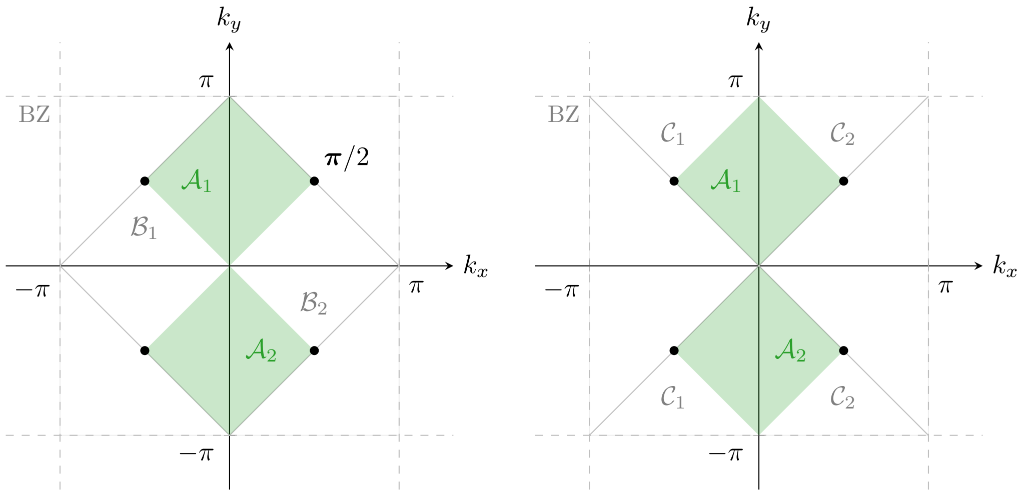
$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon^{(s^*)}(\cos k_x + \cos k_y) + \epsilon^{(d)}(\cos k_x - \cos k_y)$$

It is the balance between the two components $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ that gives the main features of the renormalised band. Consider the s^*-wave and d-wave structure factors already presented in Fig. **??**, and define the Magnetic Brillouin Zone (MBZ) and Nematic Brillouin Zone (NBZ) as follows:

$$\text{MBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \right\} \tag{1.25}$$

$$\text{NBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \tag{1.26}$$

represented in the top-left panel of Fig. 1.3. The reason for the name of the MBZ will be clear in Chap. **??**. In the rest of the panels of Fig. 1.3 are reported examples of mixed symmetry bands,



a | MBZ sectors. b | NBZ sectors.

Figure 1.4 | Depiction of the sectors composing the MBZ (Fig. 1.4a) and the NBZ (Fig. 1.4b), based on the divisions shown in the top-left panel of Fig. 1.3. The full green zone composes $\mathcal{A}$, the region where both the s^*-wave and d-wave structure factors are positive defined; its MBZ complementary white counterpart in the left panel gives $\mathcal{B}$, for which the s^*-wave structure factors is positive-definite while the d-wave is negative. Similarly the NBZ complementary white counterpart in the right panel gives $\mathcal{C}$, for which signs are exchanged. The black dots, at the contours intersection, are band nodes.

for positive components. The respective opposite sign plots are obtained by the following rules:

$$
\begin{aligned}
\epsilon^{(s^*)} < 0 \quad &\text{and} \quad \epsilon^{(d)} < 0 \quad &&\Longrightarrow \quad \text{Invert colors in Fig. 1.3} \\
\epsilon^{(s^*)} > 0 \quad &\text{and} \quad \epsilon^{(d)} < 0 \quad &&\Longrightarrow \quad \text{Rotate plots by } \pi/2 \text{ in Fig. 1.3} \\
\epsilon^{(s^*)} < 0 \quad &\text{and} \quad \epsilon^{(d)} > 0 \quad &&\Longrightarrow \quad \text{Rotate plots by } \pi/2 \text{ and invert colors in Fig. 1.3}
\end{aligned}
$$

Let us focus on the MBZ and the NBZ. Starting from the intersections of the two zones, both can be factored in two sectors of interest,

$$\mathrm{MBZ} = \mathcal{A} \cup \mathcal{B} \qquad \mathrm{NBZ} = \mathcal{A} \cup \mathcal{C}$$

being, accordingly to Fig. 1.4,

$$\mathcal{A} \equiv \mathcal{A}_1 \cup \mathcal{A}_2 \qquad \mathcal{B} \equiv \mathcal{B}_1 \cup \mathcal{B}_2 \qquad \mathcal{C} \equiv \mathcal{C}_1 \cup \mathcal{C}_2$$

These domains satisfy:

$$
\begin{aligned}
\mathcal{A} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \qquad (1.27) \\
&= \mathrm{MBZ} \cap \mathrm{NBZ}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{B} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \leq 0 \right\} \qquad (1.28) \\
&= \mathrm{MBZ} \setminus \mathrm{NBZ}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{C} &\equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \leq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \qquad (1.29) \\
&= \mathrm{NBZ} \setminus \mathrm{MBZ}
\end{aligned}
$$

and will be used later on. Whatever the shape of the mixed band, the following equation always holds

$$|k_x| = \frac{\pi}{2} \quad \text{and} \quad |k_y| = \frac{\pi}{2} \qquad \Longrightarrow \qquad \tilde{\epsilon}_{(k_x, k_y)} = 0$$

since only cosines are involved. This identifies four node points, represented as black dots in Fig. 1.4.

1.5.1 Constraints in presence of a robust C_4 symmetry and bands shrinking

Consider the simpler situation where C_4 symmetry is imposed. This implies that the renormalised bands are

$$\tilde{\epsilon}_{\mathbf{k}} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

with a single self-consistency equation

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

The following chain of deductions leads us to a constraint on the possible values of $u^{(s^*)}$:

1. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \leq 0$. Let us prove this: suppose instead $u^{(s^*)} \leq 0$. For any function $g_{\mathbf{k}}$ defined on the BZ:

   $$\sum_{\mathbf{k}\in \mathrm{BZ}} g_{\mathbf{k}} = \sum_{\mathbf{k}\in \mathrm{MBZ}} [g_{\mathbf{k}} + g_{\mathbf{k}+\boldsymbol{\pi}}]$$

   where we defined the vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. The self-consistency equation thus reads

   $$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}}; \beta, \tilde{\mu}) \right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned} \tag{1.30}$$

   having we used the antisymmetry of s^*-wave functions by $\boldsymbol{\pi}$ shifts. Now, if $u^{(s^*)} \leq 0$ the renormalised band $\tilde{\epsilon}_{\mathbf{k}}$ is negative inside the MBZ and positive outside, implying

   $$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

   the equality only being satisfied on the MBZ contour. Then, since inside the MBZ it holds by definition $\cos k_x + \cos k_y \geq 0$, the above equations imply

   $$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$

   To exclude the possibility of $u^{(s^*)} = 0$, it suffices to notice that from the argument above such a situation only arises when the MBZ contour degenerates to the entirety of the MBZ itself – which is, when the band is flat. But that cannot be the case for $u^{(s^*)} = 0$, as long as $t > 0$. Thus we have proven that

   $$u^{(s^*)} > 0$$

2. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \geq 2t/V$. We proceed exactly as did for the previous point, up to the passage:

   $$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$

   Now, if $u^{(s^*)} \geq 2t/V$ this means the prefactor in front of the renormalised bands is positive. Then $\tilde{\epsilon}_{\mathbf{k}}$ is positive inside the MBZ and negative outside, implying

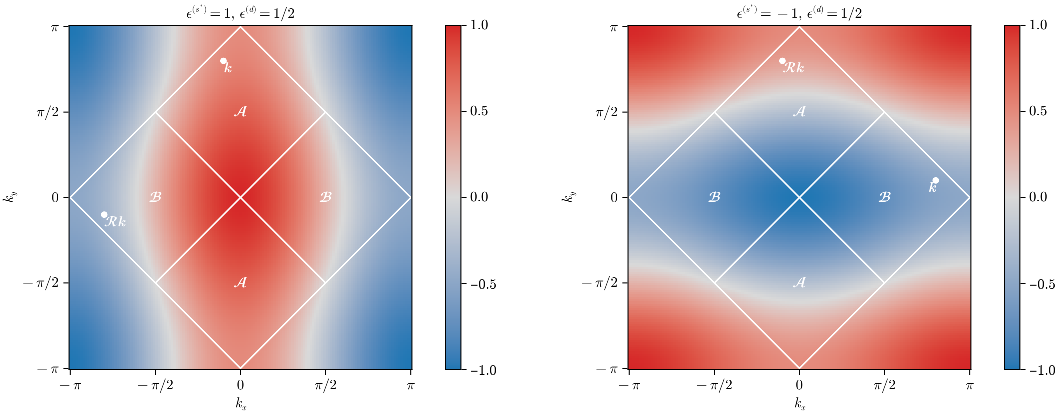
   $$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

   thus we have

   $$u^{(s^*)} \geq 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

   which is absurd. Thus

   $$u^{(s^*)} < 2t/V$$



a | Same-sign mixed bands.

b | Opposite-sign mixed bands.

Figure 1.5 | Theoretical example of bands with $u^{(s^*)} > 2t/V$, $u^{(d)} > 0$ (top-left) and $u^{(s^*)} < 0$, $u^{(d)} > 0$ (top right). Two example points are reported, connected by a $\pi/2$ rotation as given in Eqns. (1.34), (1.32).

Collecting the two points above, we have the constraint on $u^{(s^*)}$

$$\mathrm{C}_4 \quad \Longrightarrow \quad 0 < u^{(s^*)} < 2t/V \tag{1.31}$$

The effective hopping is smaller than the original hopping. We can denominate this net effect "bands shrinking".

1.5.2 Natural suppression of nematicity within $\mathrm{C}_2/\mathbb{Z}_2$ symmetry

Let us extend the arguments above to a weaker symmetry, allowing for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$$

We work with the full, anisotropic band of Eq. (1.19). From the following chain of deductions we conclude that C_4 is not broken:

1. Let for instance

$$u^{(s^*)} \geq 2t/V > 0 \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad \epsilon^{(s^*)} \geq 0 \qquad \epsilon^{(d)} > 0$$

In Fig. 1.5a an example band is shown for $\epsilon^{(s^*)} = 1$, $\epsilon^{(d)} = 1/2$. Note that if $u^{(s^*)} = 2t/V$, the renormalised band is purely d-wave since $\epsilon^{(s^*)} = 0$.

Recall Eq. (1.30), and separate in two sums over $\mathcal{A}$ and $\mathcal{B}$:

$$\begin{aligned} u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{A}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \\ + \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned}$$

Consider the sum over $\mathcal{B}$. Looking at the bottom-left panel of Fig. 1.3, such a region contains nodal curves across which bands change sign. Let $\mathcal{R}$ be the clockwise $\pi/2$ rotation: then, any point in $\mathcal{B}$ can be seen as the rotation of a point in $\mathcal{A}$

$$\forall \mathbf{k}' \in \mathcal{B} \quad \exists \quad \mathbf{k} \in \mathcal{A} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{1.32}$$

A graphical rendition of this idea is given in Fig. 1.5a. Then we can rewrite $u^{(s^*)}$ as a single sum over $\mathcal{A}$,

$$\begin{aligned} u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{A}} (\cos k_x + \cos k_y) \\ \times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned} \tag{1.33}$$

From Eq. (1.27), $\forall \mathbf{k} \in \mathcal{A}$ the form factors $\cos k_x \pm \cos k_y$ are positive by construction. By hypothesis $\epsilon^{(s^*)} \geq 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}^{\geq 0} + \overbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}_{\geq 0} - \underbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}_{\geq 0} \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \geq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \geq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

This condition, inserted in Eq. (1.33), leads to an absurd:

$$u^{(s^*)} \geq 2t/V \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad u^{(s^*)} \leq 0$$

2. An identical argument, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$, can be carried out to show that

$$u^{(s^*)} \geq 2t/V > 0 \qquad u^{(d)} < 0 \qquad \Longrightarrow \qquad u^{(s^*)} \leq 0$$

3. The two points above imply that it is possible to have $u^{(s^*)} \geq 2t/V$ if and only if $u^{(d)} = 0$. But from Sec. 1.5.1, we conclude that the upper boundary $u^{(s^*)} < 2t/V$ always holds.

4. Let for instance

$$u^{(s^*)} \leq 0 \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad \epsilon^{(s^*)} < 0 \qquad \epsilon^{(d)} > 0$$

In Fig. 1.5b an example band is shown for $\epsilon^{(s^*)} = -1$, $\epsilon^{(d)} = 1/2$.

Proceed as for the first point. Similarly to Eq. (1.32), Any point in $\mathcal{A}$ can be obtained by rotating some point in $\mathcal{B}$,

$$\forall \mathbf{k}' \in \mathcal{A} \quad \exists \quad \mathbf{k} \in \mathcal{B} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{1.34}$$

An equation similar to Eq. (1.33) holds, the sum being performed over $\mathcal{B}$

$$\begin{aligned} u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} &(\cos k_x + \cos k_y) \\ &\times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned} \tag{1.35}$$

From Eq. (1.28), $\forall \mathbf{k} \in \mathcal{B}$ the form factor $\cos k_x + \cos k_y$ is positive by construction; the form factor $\cos k_x - \cos k_y$ is negative by construction. By hypothesis $\epsilon^{(s^*)} < 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}^{\leq 0} + \overbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}^{\leq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}_{\leq 0} - \underbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}_{\leq 0} \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \leq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \leq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

This condition, inserted in Eq. (1.33), leads to an absurd:

$$u^{(s^*)} \leq 0 \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad u^{(s^*)} > 0$$

5. can be carried out to show that

$$u^{(s^*)} \leq 0 \qquad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(s^*)} > 0$$

6. The two points above imply that it is possible to have $u^{(s^*)} \leq 0$ if and only if $u^{(d)} = 0$. But from Sec. 1.5.1, we conclude that the lower boundary $u^{(s^*)} > 0$ always holds.

From the chain of derivations above, we can conclude self-consistently that also in the nematic structure the constraint of Eq. (1.31),

$$0 < u^{(s^*)} < 2t/V$$

holds. In the s^*-wave channel, partial breaking of rotational symmetry is not enough to modify the main effect identified under C_4 symmetry, "bands shrinking". What is missing is the behaviour of the d-wave channel when said effect takes place. Continuing the enumeration above,

7. Let $0 < u^{(s^*)} < 2t/V$ and $u^{(d)} > 0$. Then $\epsilon^{(d)} > 0$. The self-consistency equation for $u^{(d)}$ reads

$$\begin{aligned} u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{NBZ}} (\cos k_x - \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned}$$

where we recognised, as is shown in Fig. 1.4b, that $\mathrm{BZ} \setminus \mathrm{NBZ} = \mathcal{R}\mathrm{NBZ}$. It holds:

$$\forall \mathbf{k} \in \mathrm{NBZ} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} \end{aligned}$$

thus $\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} < \tilde{\epsilon}_{\mathbf{k}}$, and inserting this fact in the self-consistency equation above we get

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(d)} < 0$$

which is absurd.

8. An identical argument, with inverted inequalities, can be carried out to show that

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(d)} > 0$$

9. The two points above imply that it is possible to satisfy the constraint $0 < u^{(s^*)} < 2t/V$ if and only if $u^{(d)} = 0$.

All the deductions above indicate that $u^{(d)} = 0$ everywhere. Nematic effects are always absent,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \tag{1.36}$$

This is physically expected but non trivial. The normal phase does not break C_4 and we can ignore $u^{(d)}$ at all. In next section we identify the expected self-consistent value for $u^{(s^*)}$.

1.5.3 Expected dependence on δ at low temperature

We understood that $u^{(d)} = 0$. Moreover, the unique HFP $u^{(s^*)}$ is independent of U, but in general it will depend in some way on the doping level. We work at low-temperature, $\beta \to \infty$. Consider the self-consistency equation of Tab. 1.2 as a function of temperature and chemical potential,

$$u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

in the zero-temperature limit, as long as the band is not flat, it holds

$$\lim_{\beta\to\infty} u^{(s^*)}(\beta,\tilde\mu) = \frac{1}{L_xL_y}\sum_{\epsilon_{\mathbf{k}}\le\tilde\mu}(\cos k_x+\cos k_y)$$

Let FS(δ) be the Fermi surface for a given doping level δ. We denote with Int(FS) be its internal part, the Fermi sea:

$$\mathrm{FS} = \{\mathbf{k}\in \mathrm{BZ}\mid \tilde\epsilon_{\mathbf{k}} = \tilde\mu\}$$
$$\mathrm{Int(FS)} = \{\mathbf{k}\in \mathrm{BZ}\mid \tilde\epsilon_{\mathbf{k}} \le \tilde\mu\}$$

At half filling, the Fermi surface coincides with the boundary of the MBZ:

$$\delta = 0 \qquad\Longrightarrow\qquad \mathrm{FS} = \partial\mathrm{MBZ}$$

This is implied by the fact that renormalised bands are antisymmetric by $\boldsymbol{\pi}$ shifts,

$$\tilde\epsilon_{\mathbf{k}\sigma} = -\tilde\epsilon_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

The portion of an s^*-wave band outside the MBZ is mirrored with respect to the portion inside the MBZ. This implies that half the states lie in the MBZ, half out.

At $\delta>0$ the FS expands in the region BZ$\setminus$MBZ where the form factor $\cos k_x+\cos k_y$ becomes negative. This gives the self-consistency equation

$$\lim_{\beta\to\infty} u^{(s^*)}[\beta,\tilde\mu] \simeq \frac{1}{L_xL_y}\sum_{\epsilon_{\mathbf{k}}\le 0}|\cos k_x+\cos k_y| - \frac{1}{L_xL_y}\sum_{0<\epsilon_{\mathbf{k}}\le\tilde\mu}|\cos k_x+\cos k_y|$$

which, being the sum of a positive and a negative term, is maximised whenever $\tilde\mu=0$, at half filling. This holds as long as the band is not flat. Indeed, if the effective hopping is not zero, to rise the doping level means to

$$\delta_1>\delta_2 \qquad\Longrightarrow\qquad \mathrm{Int(FS_1)}\subset\mathrm{Int(FS_2)}$$

with obvious meaning of the notation. Using this property we conclude

$$\tilde t>0 \quad\text{and}\quad \delta_1>\delta_2 \qquad\Longrightarrow\qquad \lim_{\beta\to\infty}u^{(s^*)}[\beta,\tilde\mu]\Big|_{\delta_1} < \lim_{\beta\to\infty}u^{(s^*)}[\beta,\tilde\mu]\Big|_{\delta_2}$$

This is the first expected behaviour of the self-consistent functional $u^{(s^*)}$: if there exists a region of parametric space where the shrinking effect is not strong enough to flatten the band, a rise in the doping level must lower $u^{(s^*)}$. Such a region must exist: at least all the segment $(V=0,\delta)$ is included, since on said region of parametric space the model is just the common free electronic model.

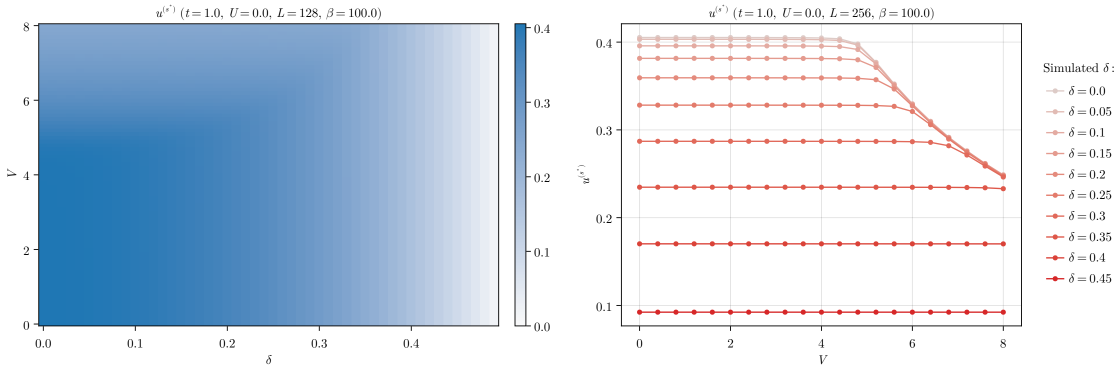
Following this line of reasoning, we can extract the two low-temperature limits of $u^{(s^*)}$ at half-filling and unitary filling.

1. At half-filling, the self-consistency equation reduces to

$$u^{(s^*)}[\beta,0]\big|_{\beta\gg1} \simeq \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}(\cos k_x+\cos k_y)$$

In thermodynamic limit, performing integration on the MBZ shown in the top-left panel of Fig. 1.3:

$$\begin{aligned}\lim_{\beta\to\infty}u^{(s^*)}[\beta,0] &\simeq \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_y\int_{-\pi+|k_y|}^{\pi-|k_y|}\mathrm{d}k_x\cos k_x + \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_x\int_{-\pi+|k_x|}^{\pi-|k_x|}\mathrm{d}k_y\cos k_y\\ &= \frac{1}{2\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\int_{-\pi-k_y}^{\pi+k_y}\mathrm{d}k_x\cos k_x + \frac{1}{2\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\int_{-\pi+k_y}^{\pi-k_y}\mathrm{d}k_x\cos k_x\\ &= \frac{1}{\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\sin(\pi+k_y) + \frac{1}{\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\sin(\pi-k_y)\\ &= \left(\frac{2}{\pi}\right)^2 \approx 0.405 \end{aligned} \tag{1.37}$$



a | Heatmap of $u^{(s^*)}$ in δV plane.

b | V-dependency of $u^{(s^*)}$ varying δ.

Figure 1.6 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase. To extract the data on the right side coarse-graining precision of the left-hand side data has been exchanged with higher lattice dimensions. As is evident, based on doping level $u^{(s^*)}$ exhibits non-trivial regions of V-independence.

2. At unitary filling we have

$$\lim_{\beta\to\infty} u^{(s^*)}\left[\beta, 2\tilde{t}^{(s^*)}\right] = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y) = 0$$

which vanishes due to the structure factor periodicity by $\boldsymbol{\pi}$.

To summarise: around $V = 0$, $u^{(s^*)}$ is expected to descend monotonically from $4/\pi^2$ to 0 at increasing doping.

1.6 Discussion of numeric results

In this section we present and discuss the numeric results, the physical relevance of the found phenomena and the boundaries of the algorithm. Let us start by sketching the algorithm used to determine self-consistently the mean-field structure of the normal phase in parameter space. For the normal phase, the HFPs "vector" defined in Sec. **??** has just one component:

$$\mathbf{v} = \left[u^{(s^*)}\right]$$

As before, we treat spin as a pure degeneracy. The only HFP is characterised by the first self-consistency equation of Tab. 1.2. The resulting model has two RMPs:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)}$$
$$\tilde{\mu} \equiv \mu + n(2zV - U)$$

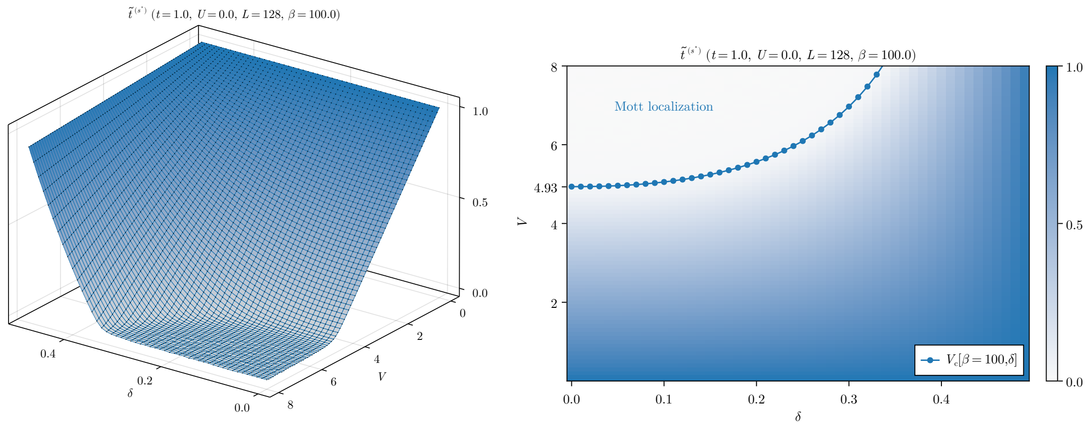
the second one being only dependent on the model parameters and not on HFPs. Finally, the phase free energy is given by Eq. (1.24). The normal phase is completely independent of U. Because of this we present the results varying the other parameters, keeping everywhere $U = 0$. We ran simulations in the (V, δ) plane, with the setup:

$$U = 0 \qquad \Delta n = 10^{-2} \qquad \Delta\mathbf{v} = \left[u^{(s^*)}: 5\times 10^{-4}\right]$$

with $\Delta\mathbf{v}$ the tolerance vector defined as in Sec. **??**.

In Fig. 1.6a are reported the results obtained for the HFP $u^{(s^*)}$ on a 128×128 lattice and "low" temperature $\beta = 100$, varying the non-local attraction and the doping level

$$V = 0,\ 0.1,\ 0.2,\ \ldots,\ 4.0 \qquad \delta = 0,\ 0.01,\ 0.02,\ \ldots,\ 0.49$$



a | Simulated values of $\tilde{t}^{(s^*)}$

b | Critical Mott boundary.

Figure 1.7 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase and the Mott-boundary V_c, computed as is described in text. In the right panel, the tick $V = 4.93$ is specified explicitly as a graphic confirm of the computation of Eq. (1.38). For graphic reasons the points at $V_c > 8$ have been omitted.

The expectations of Sec. 1.5.3 are satisfied: at low V (indicatively $V \lesssim 5$) the trend is monotonically descendent at increasing doping. As V gets bigger, however, $u^{(s^*)}$ starts decreasing in the top-left corner of Fig. 1.6a. This effect is more visible in Fig. 1.6b, where we doubled L and ran simulations with a smaller δ resolution: the convergence value of $u^{(s^*)}$ is approximately independent of V up to some point, where this behaviour is broken. Note that, as was expected from the discussion of Sec. 1.5.3, the half-filled convergence value of $u^{(s^*)}$ in the V-independent region is approximately given by Eq. (1.37).

This effect can only be interpreted in one way, as we discuss in next section: $u^{(s^*)}$ maintains a constant value at fixed doping, as is described in Sec. 1.5.3, as long as $\tilde{t}^{(s^*)} > 0$. When V gets large enough, $u^{(s^*)}$ starts decreasing because of its constraints. The net effect is the nullification of $\tilde{t}^{(s^*)}$, which is, a total localisation effect.

1.6.1 Mott localisation effect

The most interesting result we obtained for the normal phase is the possibility of localising electrons by tuning the V/δ ratio. In Fig. 1.6b, at low V the convergence value of $u^{(s^*)}$ is approximately V-independent at sufficiently low temperatures. Let $V_c[\beta, \delta]$ be the critical value such that

$$0 \leq V < V_c[\beta, \delta] \quad : \quad u^{(s^*)} \text{ is } V \text{ independent.}$$

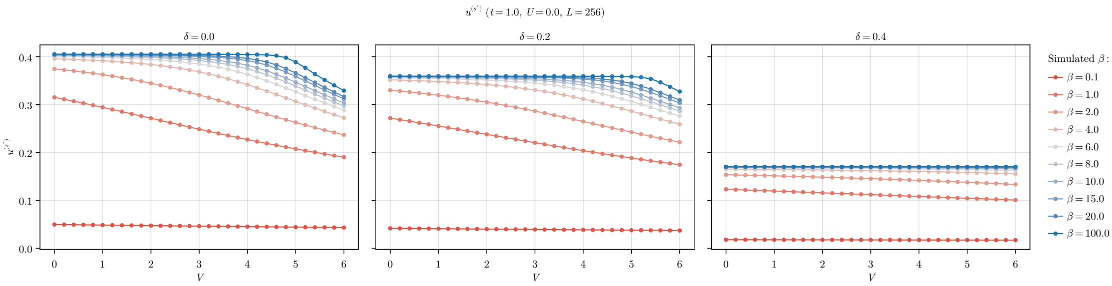
Then within the entire segment $[0, V_c]$ the convergence value is (approximately) the same obtained for $V = 0$. We can define in first instance the critical V as the value such that the hopping correction is maximized,

$$V_c[\beta, \delta] \equiv \frac{2t}{u^{(s^*)}[\beta, \tilde{\mu}(\delta)]\big|_{V=0}}$$

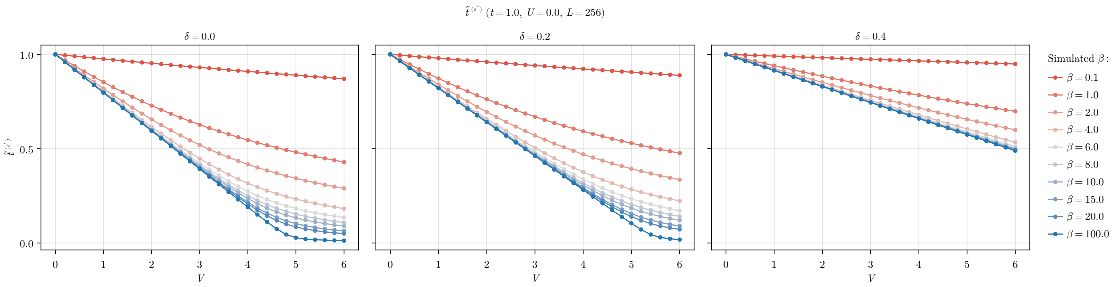
When V becomes larger than this critical value, the resulting RMP $\tilde{t}^{(s^*)}$ drops to 0 and due to the constraint of Eq. (1.31) the convergence value of $u^{(s^*)}$ is forced to decrease. Note that the constraint of Eq. (1.31) does not prevent $u^{(s^*)}$ from saturating to its upper boundary from below. In Fig. 1.7a the converged values of $\tilde{t}^{(s^*)}$, extracted from the same data of Fig. 1.6a, are plotted in parametric space.

To extract the value of $V_c[\beta, \delta]$ is a non-trivial analytic problem due to the insurgence of elliptic integrals and the smearing effects of temperature. Instead, for the high-β half-filled case we can use Eq. (1.37) and get:

$$V_c[\beta \to \infty, 0] = \frac{2t}{(2/\pi)^2} \approx 4.93t \qquad (1.38)$$



a | Temperature dependence of $u^{(s^*)}$ at various dopings.



b | Temperature dependence of $\tilde{t}^{(s^*)}$ at various dopings.

Figure 1.8 | Comparative plots of the HFP $u^{(s^*)}$ (top panel) and RMP $\tilde{t}^{(s^*)}$ (bottom panel) in null-doped (left plots), "medium-doped" ($\delta = 0.2$, central plots) and "heavily-doped" ($\delta = 0.4$, right plots) contexts. Increasing temperature progressively destroys Mott localisation, reducing the bands shrinking effect.

At half-filling, the effective bands amplitude reduction is so strong that small values of V (compared to t) are sufficient to nullify kinetics. In Fig. 1.7b we report, superimposed, the convergence values of the RMP $\tilde{t}^{(s^*)}$ and $V_c[\beta = 100, \delta]$ computed with the simple procedure hereby described. Our prediction of the critical value for V approximates reasonably the localisation region.

Increasing temperature Mott localisation is suppressed. Also, out precedent argument were valid only at low temperature. In Fig. 1.8a we report results for the HFP $u^{(s^*)}$ computed at decreasingly lower temperature and three example doping levels,

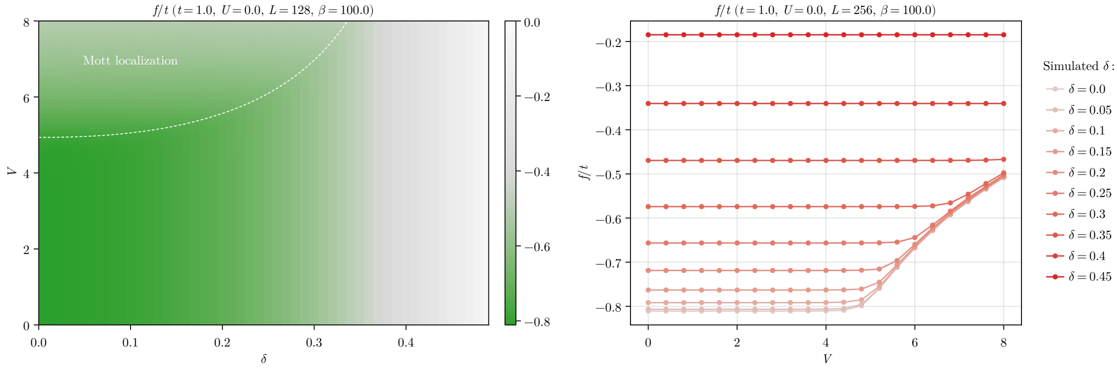
$$1.0 \leq \beta \leq 100.0 \qquad \delta = 0.0, 0.2, 0.4$$

At high temperature the V-independence region that was visible in Fig. 1.6b completely disappears, making $u^{(s^*)}$ a monotonically descendent function. In Fig. 1.8a the same dataset is plotted with focus on the RMP $\tilde{t}^{(s^*)}$. Temperature increase disrupts localisation effects, at least at small values of V. However, the bands shrinking effect is still relevant at $\beta = 1.0$, the hopping parameter approximately halving at $V \approx 5 \div 6$.

1.6.2 Solution free energy and entropy densities

In order to compare phases, we need an estimation of free energy of each. The plots of Fig. 1.9 report our findings on the normal phase free energy, computed based on Eq. (1.24). The free energy is intended in unities of t. For a non-flat band, doping increases energy because the band fills up and electrons populate higher energy levels.

Looking the data of Fig. 1.9a, at fixed δ free energy density remains pretty much constant up to the Mott boundary. This is simply because as long as a band with non-zero amplitude is present,



a | Heatmap of f in δV plane. b | V-dependency of f varying δ.

Figure 1.9 | Free energy density of the normal phase. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 1.7b, indicating the Mott transition.

the HFP correction of Eq. (1.24) compensates the free energy increase due to bands shrinking,

$$
\begin{aligned}
-V\left|u^{(s^*)}\right|^2 &= -\frac{Vu^{(s^*)*}}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \\
&= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left(\epsilon_{\mathbf{k}} - \tilde{\epsilon}_{\mathbf{k}}\right) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})
\end{aligned}
$$

having we used in the second passage

$$
\tilde{\epsilon}_{\mathbf{k}} = \tilde{\epsilon}_{\mathbf{k}} + u^{(s^*)*}V(\cos k_x + \cos k_y)
$$

Recalling the first point of the derivation of Sec. 1.4, to add the HFP correction to free energy is the same as removing the *tilde* sign from single-state energies, thus using the free model ($V = 0$) free energy density expression.

Mott localisation is energetically expensive. This is visible in both panels of Fig. 1.9. The V-dependent energy increase is an entropic effect: indeed, for a flat band:

$$
f(0;\beta,\tilde{\mu}(\delta)) = \frac{1}{2}
$$

at any temperature. The expression for entropy density of a free Fermi gas [29] is given by

$$
s(f) = -k_{\mathrm{B}}\left[f\log f + (1-f)\log(1-f)\right]
$$

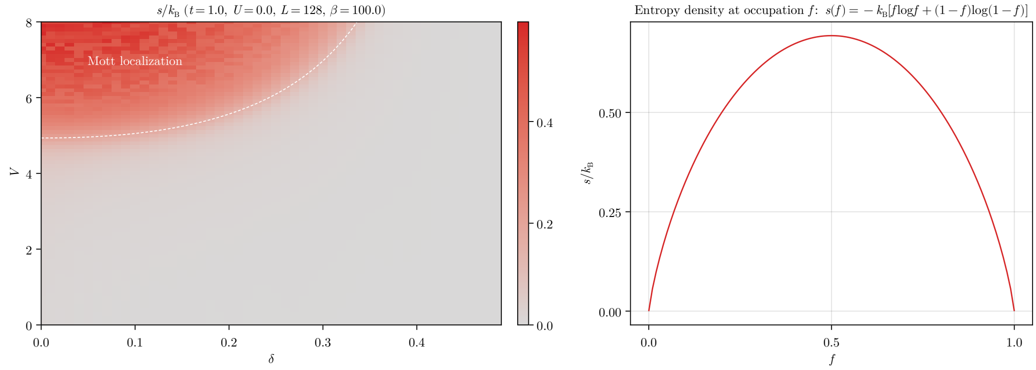
represented in Fig. 1.10b. The maximum is located at $f = 1/2$, then a flat band realises the maximally entropic condition. The Mott insulating region can be seen as a degeneration of the FS interface to the entire BZ, a process that maximises entropy.

All of this holds, as all the rest, at low temperature: by lowering β, finite-temperature effects become more and more relevant and the statements above need to be corrected.

1.6.3 Convergence problems and mixing fine-tuning

The normal phase is simple enough to describe some of the typical convergence problems arising in HF iterative schemes anticipated in Sec. **??**. Bands flattening is a source of algorithmic instability: numerics can lead to values of $u^{(s^*)}$, during iterations, slightly out of bounds with respect to Eq. (1.31). Even if the numeric effect is small, the consequence can be vast. To heal this effect and extract the actual convergence values, it is necessary to optimise the mixing parameter $g \in [0,1]$.

We defined g in Sec. **??**, and operatively in Eq. (**??**). This parameter controls deterministically how, from an initialiser (the HFPs vector we plug in the algorithm) and an output (the HFPs



a | Heatmap of s_{MFT} in δV plane.

b | V-dependency of f varying δ.

Figure 1.10 | Entropy density of the normal phase (left panel) and theoretical contribution to entropy for a state with occupation f. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 1.7b, indicating the Mott transition.

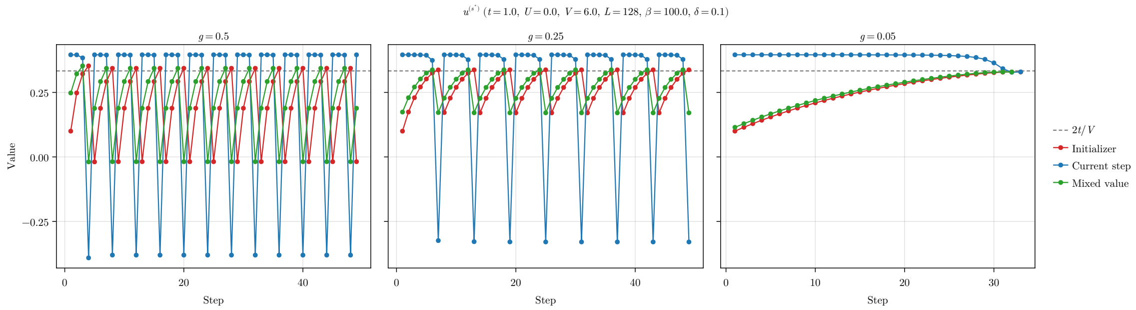


Figure 1.11 | Comparison of three example tracks for the single-valued HFPs vector of the normal phase, simulated at a “Mott-deep” point.

vector we get back from the algorithm) the new initialiser for the following step is generated. We show in Fig. 1.11 the evolutions at different mixing parameters of a simulation at

$$V = 6.0 \qquad \delta = 0.1 \qquad \beta = 100.0$$

This point is inside the Mott region, as can be checked in Fig. 1.6a. The HFP $u^{(s^*)}$ needs to converge at $2t/V = t/3$ in order to give complete localisation. The initialiser

$$\text{Step 1} \quad : \quad u^{(s^*)} = 0.1$$

was arbitrarily chosen. In each panel of Fig. 1.11, the red track represents the value of $u^{(s^*)}$ we plug in the self-consistency equations, the blue the result of the self-consistent equation, the green one the mixed value of the two based on different gs. The gray dashed line is the theoretical convergence target.

In the first two panels, there is an evident infinite loop occurring whenever the green line crosses the threshold $2t/V$. At those steps, the green point is out of bounds with respect to Eq. (1.31). When this happens, the following red point gives a band with inverted sign, a feature that “breaks” the self-consistency equation and generates a blue point with no physical significance.

To overcome this problem it is necessary to move “slowly”. As is shown in the right panel, the choice of a small enough mixing parameter ensures convergence. In the track, each blue dot is above the target and would produce a band with inverted sign. Mixing blue and red dots with a 95% unbalance on the red ones, we ensure that the green track approaches the target from below. Note, however, that lowering g cannot be reiterated too many times. Indeed, for $g \to 0$, any point of our map becomes a fixed point.

This kind of problems can turn out to be pretty common for iHF schemes, and to understand how to knob the algorithmic parameters without sacrificing accurateness and precision, as well as avoiding computationally costly and frustrating infinite loops, was an essential part of this algorithm design.

Chapter 2

Superconducting instability

In this chapter the HF methods introduced in Sec. **??** are applied to the SC phase. We study a translationally invariant superconductor, which breaks the conservation of total charge. Analytic calculations will lead us to various effects: bare bands renormalisation, anisotropic Cooper pairing and nematicity. Our description of the SC phase is closely related to the BCS solution for pure s-wave superconductivity.

As for previous phases, this chapter is divided in three parts. First, analysis of the SC phase symmetries based on the discussion of Chap. **??**, and implementation of HF methods. Second, search for theoretical constraints on the self-consistent solution. Third, numeric results.

2.1 The SC phase and its symmetries

As for antiferromagnetism, superconductivity can establish in a variety of ways. We deal with the simplest realisation. In Sec. **??** we anticipated that the SC phase is described by the two SSBs

$$\mathrm{U}^z(1) \overset{\mathrm{SSB}}{\to} 0$$
$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

Translational invariance is unbroken. Then electronic density needs to be uniform across all sites, similarly to the normal phase discussed in Chap. 1. Non-uniform density distributions are inconsistent with translational invariance in both directions. We have, then

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n \tag{2.1}$$

which is identical to Eq. (1.1) for the normal phase.

Recall Tabs. **??** and **??**, collecting the selection rules for the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. From Eqns. (2.2), (**??**) for the SC phase we need to account for the following contractions:

$\hat{H}_U$	Hartree and Cooper contractions
$\hat{H}_V$ (o.s. sector)	Hartree and Cooper contractions
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

We anticipate their roles in Tab. 2.1.

The SSB of $\mathrm{U}^z(1)$ allows for EVs like $\langle \hat{c}^\dagger \hat{c}^\dagger \rangle$ to be non-zero. This leads to the formation of Cooper pairs. These are composite quantum objects with a specific spin-spatial structure. Consider the generic Cooper fluctuation

$$\left\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\bar{\sigma}} \right\rangle \quad \text{with } i, j \text{ nearest neighbours}$$

We work in o.s. sector because we know from Tab. **??** that the s.s. sector is prohibited. From basic Quantum Mechanics we know the summation rules of the spin algebra $\mathfrak{su}(2)$,

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
		Cooper	Isotropic pairing in singlet channel
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
		Cooper	Anisotropic pairing in singlet and triplet channels
	s.s.	Hartree	μ shift by nzV
		Fock	Band renormalisation and resymmetrisation

Table 2.1 Relevant Wick's contractions and net effect in the SC phase. This RWCs table is more structured than Tabs. 1.1 and **??**.

Spatial structure	Pairing channel
Symmetric wave function	Singlet pairing
Anti-symmetric wave function	Triplet pairing

Table 2.2 Relation of the Cooper pairing channel with the wavefunction spatial symmetry (intended as the inversion $(x, y) \to (-x, -y)$).

The two pairing channels are, at this level, the singlet channel associated to total spin 0 and the triplet channel associated to total spin 1. Now, when applying HF method to the quartic operators, we get the decomposition of Eq. (**??**). We can impose to the approximated HF groundstate a particular spatial structure – say, inversion symmetry or antisymmetry. This reflects in the requirement that Cooper pairs show the same spatial structure.

This gives a selection rule over Cooper pairings: if we work with inversion-symmetric structures – say, s^*-wave or d-wave – the pairing will take place in singlet channel, allowing us to eliminate the triplet pairing. Working with antisymmetric structures, inversely, we can eliminate the singlet channel. This concept is summarised in Tab. 2.2.

We proceed as we did for the AF phase: in order to highlight the physical effects of $\hat{H}_V$, we first analyse the SC phase limitedly to the conventional HM. In next section we prove that BCS-like superconductivity is impossible at HF level on the pure HM, with just the local repulsion $\hat{H}_U$. It is the presence of the non-local attraction that enables Cooper pairing mechanism. Before starting, let

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{2.2}$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}\uparrow} \tag{2.3}$$

where $\hat{n}_{\mathbf{k}\sigma}$ was defined as for the AF phase, in Eq. (**??**). The HF approximation in reciprocal space to the EHM will be written in terms of these operators.

2.2 Local effects: (impossibility of) superconductivity in the HM

Let us start from the local effects. Following Tab. **??** and Eqns. (2.2), (**??**), we know the local repulsion decomposes in Hartree and Cooper channels,

$$\begin{aligned}
\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow} \\
&\simeq U \sum_{\mathbf{r}\in\mathcal{L}} \Big(\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle + \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle && \text{(Hartree)} \\
&\qquad + \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\langle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle + \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\rangle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} - \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \Big) && \text{(Cooper)}
\end{aligned}$$

In next sections we deal with each channel separately.

2.2.1 Local Hartree channel in the SC phase: analytic derivation

For the Hartree channel, being the density uniform across sites, we can re-use the entirety of Sec. 1.2.1 for the normal phase. Substituting Eq. (2.1), we get the approximation of Eq. (1.2),

$$\hat{H}_U \simeq nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U} + (\text{Cooper channel}) \tag{2.4}$$

thus chemical potential renormalization $\tilde{\mu} \to \mu - nU$ is present also in the SC phase.

2.2.2 Local Hartree contraction energy

As for the section above, we use the result of Sec. 1.2.2 for the normal phase. From Eq. (1.8)

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U \tag{2.5}$$

For a locally repulsive model, the Hartree channel contributes negatively to ground state energy. Indeed, contraction energy is subtracted from the hamiltonian. We now move to the Cooper channel.

2.2.3 Local Cooper channel in the SC phase: analytic derivation

We move to reciprocal space, through the transformation of Eq. (**??**). Consider the Cooper channel of Eq. (2.2). In reciprocal space we get:

$$\begin{aligned}\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \Big[&\big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+ \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle \Big]\end{aligned} \tag{2.6}$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Let us prove this: consider the EV of the Cooper pair

$$\big\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \big\rangle$$

for two momenta $\mathbf{k}_1$, $\mathbf{k}_2$. Moving to real space via an $\mathcal{F}$-transform, we get

$$\big\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \big\rangle \stackrel{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{r}_1 \in \mathcal{L}} \sum_{\mathbf{r}_2 \in \mathcal{L}} e^{-i(\mathbf{k}_1 \cdot \mathbf{r}_1 + \mathbf{k}_2 \cdot \mathbf{r}_2)} \big\langle \hat{c}^\dagger_{\mathbf{r}_1\uparrow} \hat{c}^\dagger_{\mathbf{r}_2\downarrow} \big\rangle$$

where we used the convention of Eq. (**??**). Let now $\mathbf{r}_1 \to \mathbf{r}$ and $\mathbf{r}_2 \to \Delta\mathbf{r}$. Translational invariance implies that

$$g(\Delta\mathbf{r}) = \big\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}+\Delta\mathbf{r}\downarrow} \big\rangle$$

is independent of $\mathbf{r}$. Thus we have:

$$\begin{aligned}\big\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \big\rangle &\stackrel{\mathcal{F}}{=} \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}_2 \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{-i(\mathbf{k}_1 + \mathbf{k}_2) \cdot \mathbf{r}} \\ &= \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}_2 \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \delta_{\mathbf{k}_1 = \mathbf{k}_2}\end{aligned}$$

where we used the identity of Eq. (**??**). Then the above EV vanishes for $\mathbf{k}_1 \neq \mathbf{k}_2$. Getting back to Eq. (2.6), this proves that only pairs at null total momentum can have nonzero EVs.

Let now $w_{\mathbf{k}}$ be the EV of $\phi^\dagger_{\mathbf{k}}$, and $w^{(s)}$ its s-wave projection:

$$w_{\mathbf{k}} \equiv \big\langle \phi^\dagger_{\mathbf{k}} \big\rangle \tag{2.7}$$

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} w_{\mathbf{k}} \tag{2.8}$$

Eq. (2.6) reduces to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \left[\langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \hat{\phi}_{\mathbf{k}'} + \hat{\phi}^\dagger_{\mathbf{k}} \langle \hat{\phi}_{\mathbf{k}'} \rangle \right] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{2.9}$$

being $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the HM Cooper contraction energy, written explicitly in Sec. 2.2.4. Let now

$$\Delta^{(s)} \equiv -U w^{(s)} \tag{2.10}$$

This is the s-wave component of the gap function. Note that, in general, $\Delta^{(s)} \in \mathbb{C}$. However, we can always choose $\Delta^{(s)}$ as purely real. This is due to the fact that the hamiltonian is gauge-invariant, then it suffices to apply the gauge map of Eq. (**??**) to the fermionic operators with

$$\eta \equiv \frac{\arg \Delta^{(s)}}{2}$$

to obtain $\Delta^{(s)} \in \mathbb{R}$. For the sake of generality, we proceed with $\Delta^{(s)} \in \mathbb{C}$. Although this choice gives no advantage now, we implement it anyway in order to derive the most general form of the equations that follow. This is useful for the generalisation to the EHM.

In the end, we reduce $\hat{H}_U$ from Eq. (2.9) to

$$\hat{H}_U \simeq (\text{Hartree channel}) - \sum_{\mathbf{k} \in \mathrm{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{k}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{k}} \right] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{2.11}$$

This last decomposition resembles Eq. (**??**), obtained for the AF phase in the Hartree channel.

2.2.4 Local Cooper contraction energy

The Cooper contraction energy in the Cooper channel is given by:

$$\begin{aligned} E^{(\mathrm{SC})}_{\mathrm{C}/U} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{k}\mathbf{k}'} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \langle \hat{\phi}_{\mathbf{k}'} \rangle \\ &= L_x L_y \times U \left| w^{(s)} \right|^2 \end{aligned} \tag{2.12}$$

As for the Hartree channel, also the Cooper channel contributes negatively to ground-state energy.

2.2.5 Superconductivity in the HM

We proceed here as we did for the AF phase in Sec. **??**. We start by giving a matrix representation of the approximated hamiltonian, then map it to a pseudospin problem and finally specify how to diagonalise it to retrieve its ground state.

Matrix representation. Collecting the results of the above paragraphs, we obtain the quadratic approximation to the Hubbard hamiltonian in the SC phase,

$$\begin{aligned} \hat{H}_t + \hat{H}_U \simeq & \sum_{\mathbf{k} \in \mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \\ & - \sum_{\mathbf{k} \in \mathrm{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{k}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{k}} \right] + nU \times \hat{N} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \right] \end{aligned} \tag{2.13}$$

We define the shifted bare bands:

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the renormalised chemical potential. Due to inversion symmetry, it holds $\xi_{\mathbf{k}} = \xi_{-\mathbf{k}}$. Using Fermi anticommutation rules, the kinetic operator can be written as:

$$\begin{aligned}
\hat{H}_t - \tilde{\mu}\hat{N} &= \sum_\sigma \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}\sigma}\hat{n}_{\mathbf{k}\sigma} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \left(\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}}\hat{n}_{-\mathbf{k}\downarrow}\right) \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \left(\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow} + 1\right) \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \left(\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\right) + E_{\mathrm{a}}^{(\mathrm{SC})}
\end{aligned}$$

being $E_{\mathrm{a}}^{(\mathrm{SC})}$ the "anticommutation energy"

$$E_{\mathrm{a}}^{(\mathrm{SC})} \equiv \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \tag{2.14}$$

In the end, the hamiltonian is approximated by:

$$\begin{aligned}
\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \epsilon_{\mathbf{k}} \left[\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{\mathbf{k}\downarrow}\hat{c}^\dagger_{\mathbf{k}\downarrow}\right] + E_{\mathrm{a}}^{(\mathrm{SC})} \\
- \sum_{\mathbf{q}\in\mathrm{BZ}} \left[\Delta^{(s)}\hat{\phi}_{\mathbf{q}} + \Delta^{(s)*}\hat{\phi}^\dagger_{\mathbf{q}}\right] + nU \times \hat{N} - \left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})}\right]
\end{aligned}$$

with $E_{\mathrm{H}/U}^{(\mathrm{SC})}$, $E_{\mathrm{C}/U}^{(\mathrm{SC})}$ the contraction energies arising from Hartree and Cooper channels, and $E_{\mathrm{a}}^{(\mathrm{SC})}$ the energy shift resulting from anticommutation rules, given respectively in Eqns. (2.5), (2.12), and (2.14). We recognised the s-wave gap $\Delta^{(s)}$ from Eq. (2.10).

Let the Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix}$$

The approximated hamiltonian can be written in matrix form as:

$$\hat{H}_t + \hat{H}_U - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_\sigma \hat{\Phi}^\dagger_{\mathbf{k}} h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E_{\mathrm{a}}^{(\mathrm{SC})} - \left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})}\right]$$

being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta^*_{\mathbf{k}} \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix} \quad \text{with} \quad \xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

and $\Delta_{\mathbf{k}} = \Delta^{(s)}$.

Pseudospin representation. The hamiltonian can be written equivalently as

$$h_{\mathbf{k}} = \xi_{\mathbf{k}}\tau^z - \mathrm{Re}\{\Delta_{\mathbf{k}}\}\tau^x - \mathrm{Im}\{\Delta_{\mathbf{k}}\}\tau^y$$

Let now the pseudospin operator $\hat{\mathbf{s}}_{\mathbf{k}}$

$$\hat{s}^\alpha_{\mathbf{k}} \equiv \hat{\Phi}^\dagger_{\mathbf{k}}\tau^\alpha\hat{\Phi}_{\mathbf{k}} \quad \text{with} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}\sigma} \\ \hat{s}^y_{\mathbf{k}\sigma} \\ \hat{s}^z_{\mathbf{k}\sigma} \end{bmatrix}$$

and the pseudofield $\mathbf{b}_{\mathbf{k}}$:

$$\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\mathrm{Re}\{\Delta_{\mathbf{k}}\} \\ -\mathrm{Im}\{\Delta_{\mathbf{k}}\} \\ \xi_{\mathbf{k}} \end{bmatrix}$$

EV	Meaning
$\langle \hat{s}^x_{\mathbf{k}} \rangle$	$\left\langle \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$
$\langle \hat{s}^y_{\mathbf{k}} \rangle$	$i \left\langle \hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$
$\langle \hat{s}^z_{\mathbf{k}} \rangle$	$\langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle - 1$

Table 2.3 | List of the various expectation values (EVs) of each pseudospin component and their meaning in terms of fermionic operators.

For a purely real gap, the pseudofield has no y component. In general, different gauges rotate the pseudofield around the z axis.

As was for the AF phase, these operators satisfy a $\mathfrak{su}(2)$ algebra up to an constant, irrelevant in the context of this discussion. They satisfy the identities:

$$\hat{s}^x_{\mathbf{k}} = \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \tag{2.15}$$

$$\hat{s}^y_{\mathbf{k}} = i \left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}} \right) \tag{2.16}$$

$$\hat{s}^z_{\mathbf{k}} = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow} - 1 \tag{2.17}$$

We report the EVs related to these identities Tab. 2.3.

We express the approximated hamiltonian in pseudospin representation as:

$$\hat{H}_t + \hat{H}_U - \mu \hat{N} \simeq \sum_{\mathbf{k}\in \mathrm{BZ}} \mathbf{b}_{\mathbf{k}} \cdot \hat{\mathbf{s}}_{\mathbf{k}} + nU\hat{N} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \right] \tag{2.18}$$

This operator is similar enough to the one found for the AF phase, in Eq. (??). It describes a set of pseudospin subject to a uniform field. As for the AF phase, we define the Bogoliubov bands $E_{\mathbf{k}}$

$$E_{\mathbf{k}} \equiv \sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}$$

The key difference with the AF Bogoliubov bands of Eq. (??) is that $E_{\mathbf{k}}$ now are functions of $\tilde{\mu}$, since $\xi_{\mathbf{k}}$ depends on the chemical potential. In other words, now the Bogoliubov bands depend on the shifted bands $\xi_{\mathbf{k}}$ instead of the bare bands $\epsilon_{\mathbf{k}}$.

Moreover, let us define the angles $\theta_{\mathbf{k}}$, $\zeta_{\mathbf{k}}$ as

$$\cos 2\theta_{\mathbf{k}} = \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \qquad \sin 2\theta_{\mathbf{k}} \cos 2\zeta_{\mathbf{k}} = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \qquad \sin 2\theta_{\mathbf{k}} \sin 2\zeta_{\mathbf{k}} = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}$$

Note that choosing the gauge $\eta = \arg \Delta^{(s)}/2$ the imaginary part of the gap is zero, and this reflects in $2\zeta_{\mathbf{k}} = k \times 2\pi$ with $k \in \mathbb{Z}$.

Diagonalisation. For the SC phase, diagonalisation of the HF hamiltonian follows the same procedure as was for the AF phase. Indeed, the diagonalising matrix is given by

$$\begin{aligned} W_{\mathbf{k}} &\equiv e^{-i\theta_{\mathbf{k}}\tau^y} e^{-i\zeta_{\mathbf{k}}\tau^z} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} & -\sin\theta_{\mathbf{k}} \\ \sin\theta_{\mathbf{k}} & \cos\theta_{\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{-i\zeta_{\mathbf{k}}} & \\ & e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \end{aligned} \tag{2.19}$$

and the graphic procedure is the same of Fig. ??, with $\epsilon \to \xi$. We get:

$$d_{\mathbf{k}} = W_{\mathbf{k}} h_{\mathbf{k}} W^\dagger_{\mathbf{k}} \quad \text{with} \quad d_{\mathbf{k}} \equiv \begin{bmatrix} E_{\mathbf{k}} & \\ & -E_{\mathbf{k}} \end{bmatrix} \tag{2.20}$$

In the tilted frame we have the expression for the Nambu spinors:

$$\hat{\Gamma}_{\mathbf{k}} = W_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}} \tag{2.21}$$

Explicitly:

$$
\begin{aligned}
\hat{\Gamma}_{\mathbf{k}} &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix} \\
&= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} - \sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} + \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}} \\ \hat{\gamma}^{(-)}_{\mathbf{k}} \end{bmatrix}
\end{aligned} \tag{2.22}
$$

The new Fermionic operators describe a Fermi gas on distributed on two mirrored bands,

$$
\hat{H}_t + \hat{H}_U - \tilde{\mu}\hat{N} \simeq \sum_{\mathbf{k}\in \mathrm{BZ}} \sum_{\sigma} E_{\mathbf{k}} \left(\hat{\gamma}^{(+)\dagger}_{\mathbf{k}} \hat{\gamma}^{(+)}_{\mathbf{k}} - \hat{\gamma}^{(-)\dagger}_{\mathbf{k}} \hat{\gamma}^{(-)}_{\mathbf{k}} \right) + (\cdots)
$$

with $(\cdots)$ the various energy shifts.

2.2.6 Features of the SC solution

In this section we give some details about the HF solution for the translationally-invariant SC phase. We prove that in the conventional HM, with local repulsion $\hat{H}_U$, isotropic BCS-like superconductivity is forbidden. This serves as ground to motivate the passage to the EHM.

Pseudospin EVs. The EVs of pseudospin operators in the various directions are given by:

$$
\begin{aligned}
\langle \hat{\Psi}^{\dagger}_{\mathbf{k}} \tau^x \hat{\Psi}_{\mathbf{k}} \rangle &= -\sin 2\theta_{\mathbf{k}} \cos 2\zeta_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \\
\langle \hat{\Psi}^{\dagger}_{\mathbf{k}} \tau^y \hat{\Psi}_{\mathbf{k}} \rangle &= -\sin 2\theta_{\mathbf{k}} \sin 2\zeta_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \\
\langle \hat{\Psi}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Psi}_{\mathbf{k}} \rangle &= \cos 2\theta_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle
\end{aligned}
$$

In the tilted frame, the averaged z projection of pseudospin is:

$$
\langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle = f\left(E_{\mathbf{k}}; \beta, 0\right) - f\left(-E_{\mathbf{k}}; \beta, 0\right) \tag{2.23}
$$

Compared with the respective relation for the AF phase, Eq. (**??**), the main difference here is that the chemical potential inside the Fermi-Dirac distribution is set to zero. This is due to the fact that $\tilde{\mu}$ is already included in the renormalised bands $E_{\mathbf{k}}$. The approximated quadratic hamiltonian describes a gas of γ fermions at null chemical potential.

Let us define the thermal factors:

$$
\mathrm{Th}^{(\pm)}_{\mathbf{k}}[\beta] \equiv f\left(-E_{\mathbf{k}}; \beta, 0\right) \pm f\left(E_{\mathbf{k}}; \beta, 0\right) \tag{2.24}
$$

In the end we have the EVs:

$$
\langle \hat{s}^x_{\mathbf{k}} \rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{2.25}
$$

$$
\langle \hat{s}^y_{\mathbf{k}} \rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{2.26}
$$

$$
\langle \hat{s}^z_{\mathbf{k}} \rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{2.27}
$$

Note that, as is expressed in Tab. 2.3, the z component of the pseudospin is related to density, being $1 + \langle \hat{s}^z_{\mathbf{k}} \rangle = \langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle$.

s-wave gap self-consistency equation. Applying the HF method to the conventional HM in SC phase, we obtained just one HFP: $w^{(s)}$. Let us derive its self-consistency equation. First, let

$$
\tau^{\pm} \equiv \frac{\tau^x \pm i\tau^y}{2}
$$

Recall Eq. (2.8) and expand it, to obtain

$$\begin{aligned}
w^{(s)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\langle \hat{s}^x_{\mathbf{k}} \rangle + i \langle \hat{s}^y_{\mathbf{k}} \rangle}{2} \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]
\end{aligned} \tag{2.28}$$

Last line is the explicit form of the self-consistency equation for $w^{(s)}$.

Impossibility of s-wave superconductivity. Let us define the critical U,

$$U_{\mathrm{c}}[\beta] \equiv \left[\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]}{E_{\mathbf{k}}} \right]^{-1} \tag{2.29}$$

We call it "critical" because its value gives a sort of boundary for convergence. Indeed, as we show in App. ??, for the SC phase in conventional HM, we are able to optimise parametrically the mixing parameter g in order to ensure convergence, based on the ratio $U/U_{\mathrm{c}}[\beta]$.

Let us show that superconductivity is not possible in the conventional HM with $U > 0$. Eq. (2.28) can be rewritten as:

$$w^{(s)} = -\frac{U}{U_{\mathrm{c}}[\beta]} w^{(s)}$$

By construction $U_{\mathrm{c}}[\beta] > 0$, being the thermal factor positive by construction:

$$\begin{aligned}
\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] &= \frac{1}{e^{-\beta E_{\mathbf{k}}} + 1} - \frac{1}{e^{\beta E_{\mathbf{k}}} + 1} \\
&= \frac{e^{\beta E_{\mathbf{k}}}}{e^{\beta E_{\mathbf{k}}} + 1} - \frac{1}{e^{\beta E_{\mathbf{k}}} + 1} \\
&= \tanh\left(\frac{\beta E_{\mathbf{k}}}{2} \right)
\end{aligned}$$

Hence the unique possible self-consistent solution for the equation above is $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at HF level in the repulsive HM. If instead $U < 0$, which is, if the model is attractive, we the aforementioned implication is not true and a self-consistent non-zero solution is possible.

Density. In order to estimate density, we can use the z projection of the pseudospin operators. From Eq. (2.17) we know

$$\hat{s}^z_{\mathbf{k}} + 1 = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow}$$

Then, density can be estimated as:

$$\begin{aligned}
n &= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\langle \hat{s}^z_{\mathbf{k}} \rangle + 1) \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left(1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \right)
\end{aligned} \tag{2.30}$$

We can now move to the EHM.

2.3 Extension to the EHM

Add now the non-local interaction $\hat{H}_V$. As we did for the AF phase, we look for a solution with an identical mathematical structure as the one here described for the conventional HM. We define the renormalised quantities

$$\xi_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}} \qquad E_{\mathbf{k}} \to \tilde{E}_{\mathbf{k}} \qquad \Delta_{\mathbf{k}} \to \tilde{\Delta}_{\mathbf{k}} \qquad \mu \to \tilde{\mu}$$

To be completely correct, the chemical potential was already renormalized in the HM derivation in Sec. ??. It here acquires yet another renormalization. The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}} \equiv \sqrt{\tilde{\xi}_{\mathbf{k}}^2 + |\tilde{\Delta}_{\mathbf{k}}|^2}$$

Let the renormalised angle:

$$\sin 2\tilde{\theta}_{\mathbf{k}} = \frac{|\tilde{\Delta}_{\mathbf{k}}|}{\tilde{E}_{\mathbf{k}}} \qquad \cos 2\tilde{\theta}_{\mathbf{k}} = \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tag{2.31}$$

and similarly the angle $\tilde{\zeta}_{\mathbf{k}}$,

$$\sin 2\tilde{\zeta}_{\mathbf{k}} = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \qquad \cos 2\tilde{\zeta}_{\mathbf{k}} = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \tag{2.32}$$

The pseudospin EVs are renormalised too,

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^x \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \sin 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{2.33}$$

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^y \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \cos 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{2.34}$$

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^z \hat{\Phi}_{\mathbf{k}} \rangle = \cos 2\tilde{\theta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{2.35}$$

Let also:

$$\langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle = f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) - f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \tag{2.36}$$

$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(\pm)}[\beta, \tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \pm f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) \tag{2.37}$$

For the SC phase, the thermal factors reduce to:

$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(+)} = 1 \tag{2.38}$$

$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)} = \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.39}$$

Hence, pseudospin expectation values are:

$$\langle \hat{s}_{\mathbf{k}}^x \rangle = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.40}$$

$$\langle \hat{s}_{\mathbf{k}}^y \rangle = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.41}$$

$$\langle \hat{s}_{\mathbf{k}}^z \rangle = -\frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.42}$$

These equations are an extension of Eqns. (2.40), (2.41), (2.42) for the EHM.

We can rewrite the equations for $w^{(s)}$ and n accordingly,

$$w^{(s)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{2.43}$$

$$n = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \left[1 - \cos 2\tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)\right] \tag{2.44}$$

where we used Eqns. (2.28), (2.30).

From Tabs. ?? and ?? we know none of the three channels (Hartree, Fock, Cooper) is subject to selection rules. Thus all terms of the decomposition of $\hat{H}_V$ given in Eq. (??) contribute. As always, we break down each.

2.4 Non-local Hartree effects: chemical potential renormalisation

We start by analysing the non-local Hartree channel,

$$\hat{H}_V \simeq -V \sum_{\langle ij\rangle} \sum_{\sigma,\sigma'} \Big(\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma} \langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'} \rangle + \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma} \rangle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'} - \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'} \rangle \Big)$$
$$+ \text{(Fock channel)} + \text{(Cooper channel)}$$

where we used Eq. (??). This discussion is based on that for the normal phase in Sec. 1.2.

2.4.1 Non-local Hartree channel in the SC phase: analytic derivation

Without repeating the calculations, the non-local attraction $\hat{H}_V$ is reduced in the Hartree channel to

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E^{(\text{SC})}_{\text{H}/V} + \text{(all the rest)}$$

This, together with the $\hat{H}_U$ Hartree channel decomposition

$$\hat{H}_U \simeq nU \times \hat{N} - E^{(\text{SC})}_{\text{H}/U}$$

gives the chemical potential full renormalization, identical to Eq. (1.7),

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{2.45}$$

2.4.2 Non-local Hartree contraction energy

The contraction energy for the non-local sector was derived in Eq. (1.9),

$$E^{(\text{SC})}_{\text{H}/V} = -L_x L_y \times 2zn^2 V$$

and together with the local counterpart of Eq. (1.8)

$$E^{(\text{SC})}_{\text{H}/U} = L_x L_y \times n^2 U$$

Now we move to the more interesting Fock sector.

2.5 Fock effects: bare bands renormalization

As done for the other phases, we here discuss the bands renormalization effect derived from the Fock sector of the non-local attraction. From Eq. (??), we have

$$\hat{H}_V \simeq \text{(Hartree channel)}$$
$$V \sum_{\langle ij\rangle} \sum_{\sigma,\sigma'} \Big(\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma} \rangle + \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \rangle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma} - \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \rangle \langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma} \rangle \Big) + \text{(Cooper channel)}$$

This discussion is essentially analogous the one done for the normal phase in Sec. 1.3. As for previous phases, following the selection rules of Tab. ??, only the s.s. channel contributes. Therefore we use from now on, in this section, $\sigma = \sigma'$.

2.5.1 Fock channel in the SC phase: analytic derivation

Let us define, as done for the normal phase in Eq. (1.17) and for the AF phase in Eq. (**??**),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle \tag{2.46}$$

Eq. (1.18) stands perfectly valid, translated to the SC phase

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right]$$

$$\stackrel{\mathcal{F}}{\equiv} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} u_{\mathbf{k}\sigma} \right] \times V \sum_{\mathbf{k}' \in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{k}'} \hat{c}^\dagger_{\mathbf{k}'\sigma} \hat{c}_{\mathbf{k}'\sigma} - E^{(\mathrm{SC})}_{\mathrm{F}/V} \tag{2.47}$$

being

$$E^{(\mathrm{SC})}_{\mathrm{F}/V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{k}'\sigma} \hat{c}_{\mathbf{k}'\sigma} \rangle \tag{2.48}$$

To get to this point, we employed the assumption

$$\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle$$

which essentially states translational invariance and spin degeneracy. This assumption is verified by the BCS ground state of Eq. **??**. Let us define each γ-wave component as

$$u^{(\gamma)}_{\sigma} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle$$

These are our HFPs in the Fock channel. The SC phase does not break inversion symmetry: then, as we concluded for the other phases, p_ℓ components are null. The renormalised bands are, as in Eqns. (1.19), (**??**),

$$\tilde{\epsilon}_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} + u^{(s^*)} V (\cos k_x + \cos k_y) + u^{(d)} V (\cos k_x - \cos k_y)$$

Let us compute the EVs of $u_{\mathbf{k}\sigma}$ in the two spin sectors,

$$\begin{aligned} u_{\mathbf{k}\uparrow} &= \langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}_{\mathbf{k}\uparrow} \rangle \\ &= \left\langle \hat{\Phi}^\dagger_{\mathbf{k}} \frac{\mathbb{I} + \tau^z}{2} \hat{\Phi}_{\mathbf{k}} \right\rangle \\ &= \frac{1}{2} \left[\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}} \hat{\Gamma}_{\mathbf{k}} \right\rangle + \langle \hat{s}^z_{\mathbf{k}} \rangle \right] \\ &= \frac{1}{2} \left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta] - \cos 2\tilde{\theta}_{\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta] \right] \\ &= \frac{1}{2} \left[1 - \cos 2\tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2} \right) \right] \end{aligned} \tag{2.49}$$

having we used Eq. (**??**) and following. For the spin $\downarrow$ EV,

$$\begin{aligned} u_{-\mathbf{k}\downarrow} &= \langle \hat{c}^\dagger_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow} \rangle \\ &= 1 - \langle \hat{c}_{-\mathbf{k}\downarrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \rangle \\ &= 1 - \left\langle \hat{\Phi}^\dagger_{\mathbf{k}} \frac{\mathbb{I} - \tau^z}{2} \hat{\Phi}_{\mathbf{k}} \right\rangle \\ &= 1 - \frac{1}{2} \left[\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}} \hat{\Gamma}_{\mathbf{k}} \right\rangle - \langle \hat{s}^z_{\mathbf{k}} \rangle \right] \\ &= 1 - \frac{1}{2} \left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta] + \cos 2\tilde{\theta}_{\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta] \right] \\ &= 1 - \frac{1}{2} \left[1 + \cos 2\tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2} \right) \right] \end{aligned} \tag{2.50}$$

Symmetry	**Self-consistency equation**	**Graph**
s^*-wave	$u^{(s^*)} = \dfrac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y)\left[1 - \dfrac{\tilde{\xi}_\mathbf{k}}{\tilde{E}_\mathbf{k}}\tanh\left(\dfrac{\beta\tilde{E}_\mathbf{k}}{2}\right)\right]$	Fig. **??**
$d_{x^2-y^2}$-wave	$u^{(d)} = \dfrac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x - \cos k_y)\left[1 - \dfrac{\tilde{\xi}_\mathbf{k}}{\tilde{E}_\mathbf{k}}\tanh\left(\dfrac{\beta\tilde{E}_\mathbf{k}}{2}\right)\right]$	Fig. **??**

Table 2.4 | Symmetry resolved self-consistency equations for the harmonics of $u_\mathbf{k}$.

From the above results, note first that

$$u_{\mathbf{k}\uparrow} = u_{-\mathbf{k}\downarrow}$$

which is coherent with our requirement of a uniform-density, spin uniform and inversion symmetric phase. Note also that Eqns. (2.49), (2.50) combine to give the expression for density of Eq. (2.44).

If we take the $\tilde{\theta}_\mathbf{k} \to 0$ limit, we must retrieve the Fermi-Dirac distribution. Indeed, this limit coincides with $U \to 0$. Explicitly:

$$\begin{aligned}
\lim_{\tilde{\theta}_\mathbf{k}\to 0} u_{\mathbf{k}\sigma} &= \frac{1}{2}\left[1 + \tanh\left(\frac{\beta\tilde{\xi}_\mathbf{k}}{2}\right)\right] \\
&= \frac{1}{2}\left[\frac{e^{\beta\tilde{\xi}_\mathbf{k}/2} + e^{-\beta\tilde{\xi}_\mathbf{k}/2}}{e^{\beta\tilde{\xi}_\mathbf{k}/2} + e^{-\beta\tilde{\xi}_\mathbf{k}/2}} - \frac{e^{\beta\tilde{\xi}_\mathbf{k}/2} - e^{-\beta\tilde{\xi}_\mathbf{k}/2}}{e^{\beta\tilde{\xi}_\mathbf{k}/2} + e^{-\beta\tilde{\xi}_\mathbf{k}/2}}\right] \\
&= f(\tilde{\epsilon}_\mathbf{k}; \beta, \tilde{\mu})
\end{aligned}$$

Correctly, the null-gap limit (for which the angle reduces to zero) gives back a perfectly degenerate Fermi gas.

We can then derive the $u_\mathbf{k}$-related HFPs self-consistency equations. In next passages, we use $\sigma = \uparrow$ but could have used the opposite due to spin degeneracy discussed above. Then:

$$\begin{aligned}
u^{(\gamma)} &\equiv \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_\mathbf{k}^{(\gamma)}\langle\hat{n}_{\mathbf{k}\uparrow}\rangle \\
&= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_\mathbf{k}^{(\gamma)}\left[1 - \cos 2\tilde{\theta}_\mathbf{k}\tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right)\right]
\end{aligned} \tag{2.51}$$

reported explicitly for the only two inversion symmetric structures in Tab. 2.4.

2.5.2 Fock contraction energy

In terms of the energy shift due to Wick's contraction, Eq. (1.20) holds identically

$$E_{\mathrm{F}/V}^{(\mathrm{SC})} = L_xL_y \times V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right]$$

with the only obvious difference that the $u^{(\gamma)}$ are computed on the SC ground state by the means of Eqns. (2.49), (2.50).

2.6 Non-local Cooper effects: anisotropic pairing

In this section we deal with the Cooper part of Eq. (**??**),

$$\begin{aligned}
\hat{H}_V =\simeq\ &(\text{Hartree channel}) + (\text{Fock channel}) \\
&- V\sum_{\langle ij\rangle}\sum_{\sigma,\sigma'}\left(\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\langle\hat{c}_{j\sigma'}\hat{c}_{i\sigma}\rangle + \langle\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\rangle\hat{c}_{j\sigma'}\hat{c}_{i\sigma} - \langle\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\rangle\langle\hat{c}_{j\sigma'}\hat{c}_{i\sigma}\rangle\right)
\end{aligned}$$

We know from Eq. (**??**) that the non-local attraction can be separated in o.s. and s.s. channels. As we reported in Tab. **??** and stated in Sec. 2.1, the only non-zero contribution comes from the o.s. sector.

2.6.1 Non-local Cooper o.s. channel in the SC phase: analytic derivation

The o.s. Cooper sector contributes both to singlet and triplet pairing. Note that in singlet pairing channel the spatial wavefunction must be inversion-symmetric. On the contrary, in triplet pairing channel it needs to be inversion-antisymmetric.

We move to reciprocal space. From Eq. (**??**), we get

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \times \Big[\big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle \Big]$$

Identical considerations as for the local repulsion hold, and uniquely the $\mathbf{K} = \mathbf{0}$ term gives a non-zero contribution, in general. Therefore:

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - \frac{2V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \times \Big[\big\langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \big\rangle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \big\langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \big\rangle \Big]$$

Then, using the harmonics decomposition of Eq. (**??**), we end up with:

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - \frac{V}{L_x L_y} \sum_\gamma \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} \left[w_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} + w^*_{\mathbf{k}} \hat{\phi}^\dagger_{\mathbf{k}'} \right] - E^{(\text{SC})}_{\text{C}/V/\text{o.s.}} \tag{2.52}$$

In the equation above the contraction energy is given by

$$E^{(\text{SC})}_{\text{C}/V/\text{o.s.}} \equiv -\frac{V}{L_x L_y} \sum_\gamma \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} w_{\mathbf{k}} w^*_{\mathbf{k}'} \tag{2.53}$$

Then, let the harmonics $w^{(\gamma)}$ be

$$w^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} w_{\mathbf{k}}$$

Eq. (2.52) becomes

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - V \sum_\gamma \sum_{\mathbf{k}\in\text{BZ}} \left[w^{(\gamma)} \varphi^{(\gamma)*}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}} + \text{h.c.} \right] - E^{(\text{SC})}_{\text{C}/V/\text{o.s.}}$$

extending the symmetry sum to $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$.

For the γ-wave component of the superconducting order parameter, the following chain, similar to Eq. (2.28), holds:

$$\begin{aligned} w^{(\gamma)} &\equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\langle \hat{s}^x_{\mathbf{k}} \rangle + i \langle \hat{s}^y_{\mathbf{k}} \rangle}{2} \\ &= \frac{1}{2 L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \text{Th}^{(-)}_{\mathbf{k}}[\beta] \end{aligned} \tag{2.54}$$

Symmetry	Self-consistency equation	Graph
s-wave	$w^{(s)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
s^*-wave	$w^{(s^*)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y)\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
p_x-wave	$w^{(p_x)} = \frac{i\sqrt{2}}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_x \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
p_y-wave	$w^{(p_y)} = \frac{i\sqrt{2}}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_y \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
$d_{x^2-y^2}$-wave	$w^{(d)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y)\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??

Table 2.5 | Symmetry resolved self-consistency equations for the harmonics of $w_{\mathbf{k}}$.

where $\tau^+ = (\tau^x + i\tau^y)/2$. We used Eqns. (2.33), (2.34) and (2.36) as well as the thermal factor defined in Eq. (2.24). The derivation above holds for all symmetries, (including s-wave). Explicitly, five self-consistency equations are then derived and listed in Tab. 2.5.

Of these, the singlet channel admits for non-zero fluctuations of the superconducting order parameter uniquely in the s^* and d channels. This is because the gap function is inversion-symmetric and p_ℓ channels equations integrate it with an antisymmetric function.

2.7 Full MFT hamiltonian and anisotropic pairing

Let now the total contraction energy

$$E_{\mathrm{c}}^{(\mathrm{SC})} \equiv -\left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})} + E_{\mathrm{H}/V}^{(\mathrm{SC})} + E_{\mathrm{F}/V}^{(\mathrm{SC})} + E_{\mathrm{C}/V/\mathrm{o.s.}}^{(\mathrm{SC})}\right]$$

By composing all three channels we are able to derive the final form of the MFT gran-canonical hamiltonian,

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}}\left[\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} + \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\right] - \sum_{\mathbf{k}\in\mathrm{BZ}}\left[\tilde{\Delta}_{\mathbf{k}}\hat{\phi}_{\mathbf{k}} + \tilde{\Delta}^*_{\mathbf{k}}\hat{\phi}^\dagger_{\mathbf{k}}\right] + E_{\mathrm{a}}^{(\mathrm{SC})} + E_{\mathrm{c}}^{(\mathrm{SC})}$$

having we included the "anticommutation energy" also defined for the conventional HM in Eq. (??),

$$E_{\mathrm{a}}^{(\mathrm{SC})} \equiv \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}}$$

coming from fermionic anticommutation relations that are employed in order to get the kinetic term. The renormalized bands are

$$\tilde{\xi}_{\mathbf{k}} = \epsilon_{\mathbf{k}} + V\sum_{\gamma\in\{s^*,d\}} u^{(\gamma)}\varphi^{(\gamma)*}_{\mathbf{k}} - \tilde{\mu} \tag{2.55}$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}} = -Uw^{(s)} + V\sum_{\gamma} w^{(\gamma)}\varphi^{(\gamma)*}_{\mathbf{k}} \tag{2.56}$$

We can identify the gap harmonics by

$$\tilde{\Delta}^{(s)} = -Uw^{(s)} \qquad \tilde{\Delta}^{(\gamma)} = Vw^{(\gamma)} \quad \text{being} \quad \gamma\in\{s^*, p_x, p_y, d\}$$

As anticipated, the singlet channel admits three components only,

$$\text{Singlet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = \tilde{\Delta}^{(s)} + \tilde{\Delta}^{(s^*)}(\cos k_x + \cos k_y) + \tilde{\Delta}^{(d)}(\cos k_x - \cos k_y)$$

making the gap function purely real. An inverse statement holds for the triplet pairing, for which

$$\text{Triplet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = i\sqrt{2}\left[\tilde{\Delta}^{(p_x)} \sin k_x + \tilde{\Delta}^{(p_y)} \sin k_y\right]$$

In this case, the gap is purely imaginary. When working in a specific channel, the gap function is either purely real or purely imaginary. This lets us discuss the shape of the renormalised ground-state.

2.8 The zero-temperature ground-state at half-filling

The HF approximation to the true ground-state in the SC phase is obtained by aligning the each pseudospin with the respective pseudo-field. Let the up and down states,

$$|\Uparrow_{\mathbf{k}}\rangle \qquad |\Downarrow_{\mathbf{k}}\rangle$$

Using now Eq. (2.23), we have

$$\begin{aligned}\langle\Downarrow_{\mathbf{k}}|\hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}}|\Downarrow_{\mathbf{k}}\rangle &= 0\\ \langle\Uparrow_{\mathbf{k}}|\hat{s}^z_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}}|\Uparrow_{\mathbf{k}}\rangle &= 2\end{aligned}$$

so up to a phase

$$|\Downarrow_{\mathbf{k}}\rangle \sim |\Omega_{\mathbf{k}}\rangle \qquad |\Uparrow_{\mathbf{k}}\rangle \sim \hat{\phi}^\dagger_{\mathbf{k}}\,|\Omega_{\mathbf{k}}\rangle$$

The derivations carried out for the AF phase in Sec. **??** can be re-used. We end up with the state:

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}}\left[\cos\tilde{\theta}_{\mathbf{k}} + e^{2i\tilde{\zeta}_{\mathbf{k}}}\sin\tilde{\theta}_{\mathbf{k}}\hat{\phi}^\dagger_{\mathbf{k}}\right]|\Omega_{\mathbf{k}}\rangle$$

which is an extension of the conventional BCS state. The phase factor angle is the gap "tilt" in the complex plane. This is proven by:

$$\begin{aligned}\langle\Psi|\hat{s}^x_{\mathbf{k}}|\Psi\rangle &= \langle\Psi|\hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}}|\Psi\rangle\\ &= \sin\tilde{\theta}_{\mathbf{k}}\cos\tilde{\theta}_{\mathbf{k}}\left(e^{2i\tilde{\zeta}_{\mathbf{k}}} + e^{-2i\tilde{\zeta}_{\mathbf{k}}}\right)\\ &= \sin(2\tilde{\theta}_{\mathbf{k}})\cos(2\tilde{\zeta}_{\mathbf{k}})\end{aligned}$$

$$\begin{aligned}\langle\Psi|\hat{s}^y_{\mathbf{k}}|\Psi\rangle &= i\,\langle\Psi|\hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}}|\Psi\rangle\\ &= i\sin\tilde{\theta}_{\mathbf{k}}\cos\tilde{\theta}_{\mathbf{k}}\left(e^{-2i\tilde{\zeta}_{\mathbf{k}}} - e^{2i\tilde{\zeta}_{\mathbf{k}}}\right)\\ &= \sin(2\tilde{\theta}_{\mathbf{k}})\sin(2\tilde{\zeta}_{\mathbf{k}})\end{aligned}$$

These results are coherent with Eqns. (2.33) as well as (2.34) and allow us to identify the phase with angle of the pseudofield in the xy plane.

2.9 SC free energy density

Free energy density of the SC phase is derived following the procedure used for the other two phases. As for the AF phase of Sec. **??**, two fermionic bands are present; however, recalling the antiferromagnet entropy density of Eq. (**??**), the key differences are here given by the fact that the chemical potential inside the Fermi-Dirac factors is null and the sum is extended on the full BZ (but not later on spin index),

$$\begin{aligned}s^{(\pm)} \equiv -\frac{k_B}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\Big[\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\Big)\ln\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\Big)\\ + f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\ln f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\Big]\end{aligned} \tag{2.57}$$

with identical meaning of the notation. Using the relation

$$\frac{1}{e^{x}+1}+\frac{1}{e^{-x}+1}=1$$

we conclude that

$$s^{(-)}=s^{(+)} \quad\Longrightarrow\quad s\equiv s^{(-)}+s^{(+)}=2s^{(+)}$$

being s the total entropy density. The five contributions to free energy are:

1. Quasiparticles energy:
$$\begin{aligned} e_{\mathrm{g}}^{(\mathrm{SC})} &= -\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{E}_{\mathbf{k}}\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta] \\ &= -\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{E}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) \end{aligned}$$

2. Anticommutation energy:
$$\begin{aligned} e_{\mathrm{a}}^{(\mathrm{SC})} &\equiv \frac{E_{\mathrm{a}}^{(\mathrm{SC})}}{L_x L_y} \\ &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{\xi}_{\mathbf{k}} \end{aligned}$$

3. Contraction energy:
$$e_{\mathrm{c}}^{(\mathrm{SC})} = -U\left[n^2+\left|w^{(s)}\right|^2\right]+V\left[2zn^2-\sum_{\gamma}\left|u^{(\gamma)}\right|^2+\sum_{\gamma}\left|w^{(\gamma)}\right|^2\right]$$

4. Entropic energy
$$e_{s}^{(\mathrm{SC})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big)-\frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{E}_{\mathbf{k}}f(\tilde{E}_{\mathbf{k}};\beta,0)$$

5. Chemical potential correction
$$e_{n}^{(\mathrm{SC})}=\mu n$$

Compose everything

$$f^{(\mathrm{SC})}\equiv e_{\mathrm{g}}^{(\mathrm{SC})}+e_{\mathrm{a}}^{(\mathrm{SC})}+e_{\mathrm{c}}^{(\mathrm{SC})}+e_{s}^{(\mathrm{SC})}+e_{n}^{(\mathrm{SC})}$$

and use that:

$$\tanh\left(\frac{x}{2}\right)+\frac{2}{e^{x}+1}=1$$

In the end we get:

$$\begin{aligned} f^{(\mathrm{SC})} = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[\tilde{\xi}_{\mathbf{k}}-\tilde{E}_{\mathbf{k}}\right]+\frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \\ -U\left[n^2+\left|w^{(s)}\right|^2\right]+V\left[2zn^2-\sum_{\gamma}\left|u^{(\gamma)}\right|^2+\sum_{\gamma}\left|w^{(\gamma)}\right|^2\right]+\mu n \end{aligned}$$

As was for the normal and AF phases, we recognise the chemical potential shift of Eq. (2.45),

$$\begin{aligned} f^{(\mathrm{SC})} = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[\tilde{\xi}_{\mathbf{k}}-\tilde{E}_{\mathbf{k}}\right]+\frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \\ -U\left|w^{(s)}\right|^2-V\left[\sum_{\gamma}\left|u^{(\gamma)}\right|^2-\sum_{\gamma}\left|w^{(\gamma)}\right|^2\right]+\tilde{\mu}n \end{aligned} \tag{2.58}$$

which gives the analytic expression for the free energy.

2.9.1 The gap energy

Similarly to Sec. ??, let us prove the following equation about the gap energy:

$$-U\left|w^{(s)}\right|^2+V\sum_\gamma\left|w^{(\gamma)}\right|^2=\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{\left|\tilde{\Delta}_\mathbf{k}\right|^2}{\tilde{E}_\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right) \tag{2.59}$$

Expand the right-hand side through Eq. (2.56),

$$\begin{aligned}\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{\left|\tilde{\Delta}_\mathbf{k}\right|^2}{\tilde{E}_\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right)&=\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[-Uw^{(s)*}+V\sum_\gamma w^{(\gamma)*}\varphi_\mathbf{k}^{(\gamma)}\right]\frac{\tilde{\Delta}_\mathbf{k}}{\tilde{E}_\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right)\\&=-Uw^{(s)*}\times\overbrace{\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{\tilde{\Delta}_\mathbf{k}}{\tilde{E}_\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right)}^{w^{(s)}}\\&\quad+V\sum_\gamma w^{(\gamma)*}\times\underbrace{\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi_\mathbf{k}^{(\gamma)}\frac{\tilde{\Delta}_\mathbf{k}}{\tilde{E}_\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right)}_{w^{(\gamma)}}\end{aligned}$$

having we used the self-consistency equations of Tab. 2.5. This proves Eq.(2.59), which is similar to Eq. (??) with one exception. Since the right-hand side of Eq.(2.59) is positive, for $V>0$ we have the constraint

$$\sum_\gamma\left|w^{(\gamma)}\right|^2\geq M^{(\mathrm{SC})}\quad\text{being}\quad M^{(\mathrm{SC})}\equiv\left|w^{(s)}\right|^2\frac{U}{V} \tag{2.60}$$

Now, as we state in Sec. 2.2.6, for the pure HM we have identically $w^{(s)}=0$. The pure HM cannot exhibit BCS-like superconductivity. For the EHM instead we see that:

- In the strong coupling limit $U\ll V$, even small s-wave gaps impose larger $\gamma\neq s$ components;
- The s-wave gap $w^{(s)}$ cannot converge to a finite value if for any $\gamma\neq s$ we have $w^{(\gamma)}=0$.

Of course, Eq. (2.59) is also solved by a null gap on both sides: this indicates that when the constraint is impossible to fulfil self-consistently, the gap closes.

2.10 Discussion of numeric results for the singlet channel

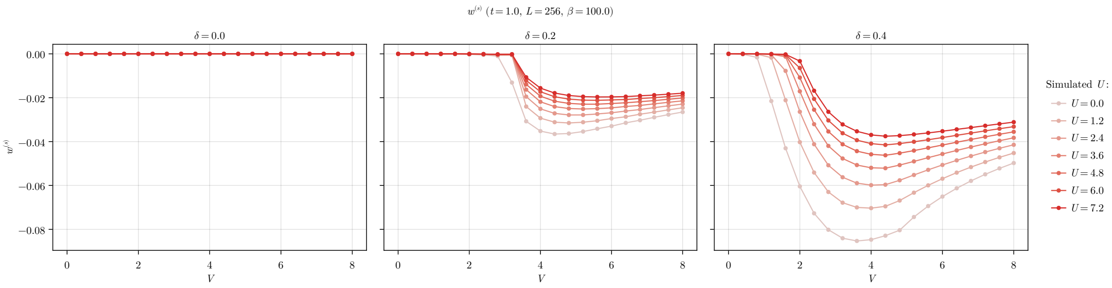
The last part of this chapter is devoted to presenting and discussing the numeric results we extracted from the iHF procedure. **We limited simulations to the singlet channel, because** [I wanted to see the end of this project!]. We specify that, differently from AF and normal phases, for the singlet SC phase we did not implement any theoretical constraint on the five HFPs,

$$\mathbf{v}=\begin{bmatrix}u^{(s^*)}\\u^{(d)}\\w^{(s)}\\w^{(s^*)}\\w^{(d)}\end{bmatrix}\qquad\mathbf{v}=\mathbb{R}^5$$

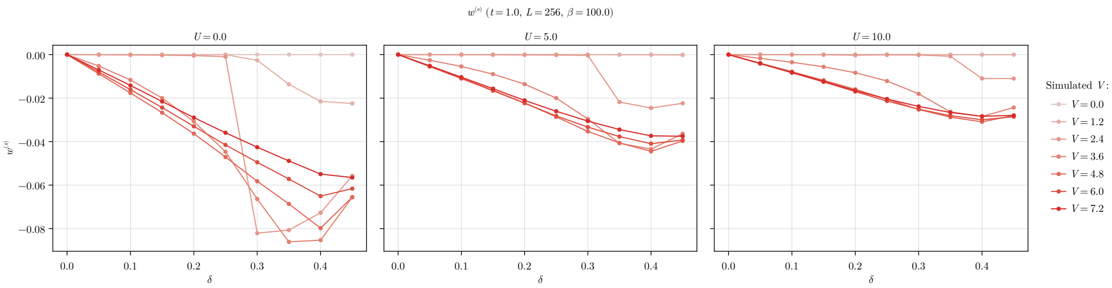
As we will discuss, all five HFPs are significantly non-zero in this phase, at least in some portion of parameters space. Therefore, we could at most search theoretical constraints describing the boundaries of these portions of space. We did not, and just ran unconstrained simulation by brute-force.

The results we obtained indicate that this phase is described by three RMPs:

$$\begin{aligned}\tilde{t}_{ij}^{(s^*)}&\equiv\left(t-\frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)}\\\tilde{t}_{ij}^{(d)}&\equiv-\frac{u^{(s^*)}V}{2}\varphi_{ij}^{(d)}\\\tilde{\mu}&\equiv\mu+n(2zV-U)\end{aligned}$$



a | Dependency of $w^{(s)}$ on V at various Us, compared at three values of δ.



b | Dependency of $w^{(s)}$ on δ at various Vs, compared at three values of U.

Figure 2.1 | Dependency of the s-wave HFP $w^{(s)}$ on V at fixed U and δ (top panel, compared figures) and on δ at fixed V and U (bottom panel, compared figures).

For AF and normal phase, we had two HFPs and no d-wave component of the renormalised hopping. We ran simulations at $\beta = 100.0$ within the boundaries:

$$0 \leq U \leq 12 \qquad 0 \leq V \leq 8 \qquad 0 \leq \delta \leq 0.45$$

For the normal phase we did not vary U; for the AF phase we did not vary δ. Here, we need to vary both in order to explore the entire parametric space.

2.10.1 Multi-symmetry singlet superconductivity

In singlet channel, the most general gap function is a combination of s, s^* and d components. The resulting gap function $\tilde{\Delta}_{\mathbf{k}}$ emerges from a competition among these channels. Running unconstrained simulations, we are able to see directly the result of the competition in the self-consistent solution. We discuss the three channels separately.

s-wave gap. Let us discuss the local isotropic harmonic of the gap function, namely the s-wave component. In Fig. 2.1, we show the data for $w^{(s)}$. The isotropic local gap is given by:

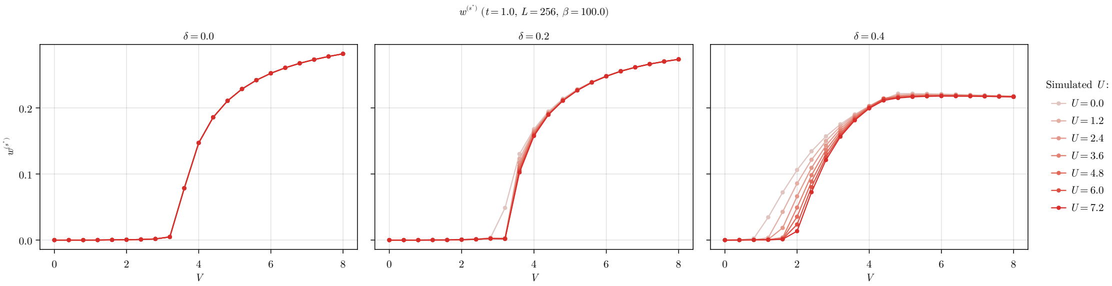
$$\tilde{\Delta}^{(s)} = -U w^{(s)}$$

In the conventional HM we know that s-wave superconductivity is impossible at HF level. This was discussed in Sec. (2.2.5). Instead in the EHM there is no selection rule for $w^{(s)}$. The only constraint comes from Eq. (2.60): $w^{(s)}$ can be non-zero as long as at least one $w^{(\gamma)}$ with $\gamma = s^*, d$ is non-zero. This is a necessary but insufficient condition.

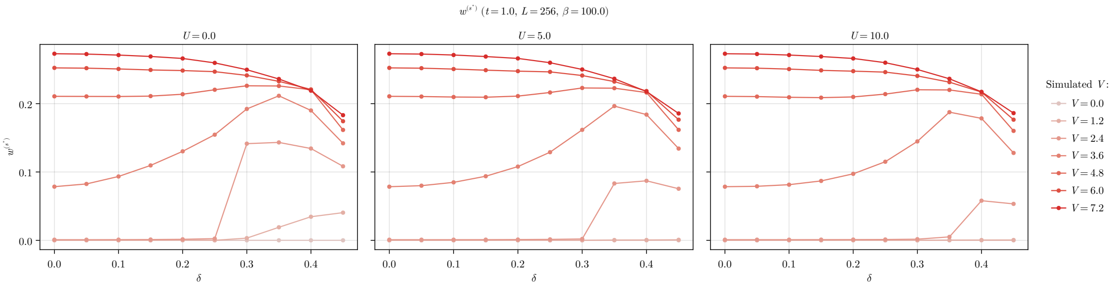
In Fig. 2.1a we see that the local isotropic gap HFP $w^{(s)}$ is null at $\delta = 0$ (left panel). Hence

$$\forall\ U/t, V/t \quad : \quad \tilde{\Delta}^{(s)}\big|_{\delta=0} = 0 \tag{2.61}$$

at least within the range of parameters we explored. Increasing doping (central and right panels), $w^{(s)}$ is null up to a critical value $V_c^{(s)}$, which depends also on U. For $V \geq V_c^{(s)}$, $w^{(s)}$ is negative and the shape of the curve depends strongly on U. In particular, the amplitude $\left|w^{(s)}\right|$ exhibits a local



a | Dependency of $w^{(s^*)}$ on V at various Us, compared at three values of δ.



b | Dependency of $w^{(s^*)}$ on δ at various Vs, compared at three values of U.

Figure 2.2 | Dependency of the s^*-wave HFP $w^{(s^*)}$ on V at fixed U and δ (top panel, compared figures) and on δ at fixed V and U (bottom panel, compared figures).

maximum. At almost unitary filling, ratios V/t comparable with 1 are sufficient to develop a finite convergence value for $w^{(s)}$. Therefore, doping lowers the critical value of the non-local attraction that is needed in order to establish an isotropic gap in the local channel.

The dependence of $w^{(s)}$ on doping is shown in Fig. 2.1b. As anticipated, $w^{(s)} = 0$ at half-filling. Varying U/t and V/t we see that $w^{(s)} = 0$ reaches some negative minimum whose amplitude is lower at higher Us.

s^*-wave gap. The results for the non-local isotropic HFP $w^{(s^*)}$ are shown in Fig. 2.2. This component of the full gap is given by

$$\tilde{\Delta}^{(s^*)} = V w^{(s^*)}$$

Consider first Fig. 2.2a. At null and finite fillings, $w^{(s^*)}$ is null up to a critical value $V_{\mathrm{c}}^{(s^*)}$. Its position depends on δ and weakly on U/t. The half-filling curve (left panel) is independent of U/t. Differently from $w^{(s)}$, we have

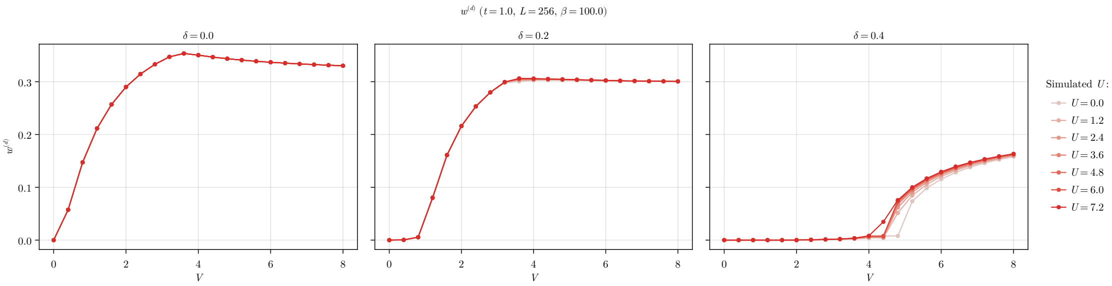
$$\forall\; U/t \quad \wedge \quad \forall V < V_{\mathrm{c}}^{(s^*)} \quad : \quad \tilde{\Delta}^{(s)}\big|_{\delta=0} = 0$$

The local isotropic gap harmonic $\tilde{\Delta}^{(s)}$ was always null at half filling. Its non-local counterpart $\tilde{\Delta}^{(s^*)}$ is null up to the critical value delimiting the phase transition. Therefore, superconductivity can establish at half-filling with a gap whose isotropic part is purely non-local. Note that we are not saying that pure s^*-wave superconductivity is possible. As we discuss later, at half-filling the dominant harmonic of the gap function has d-wave symmetry.

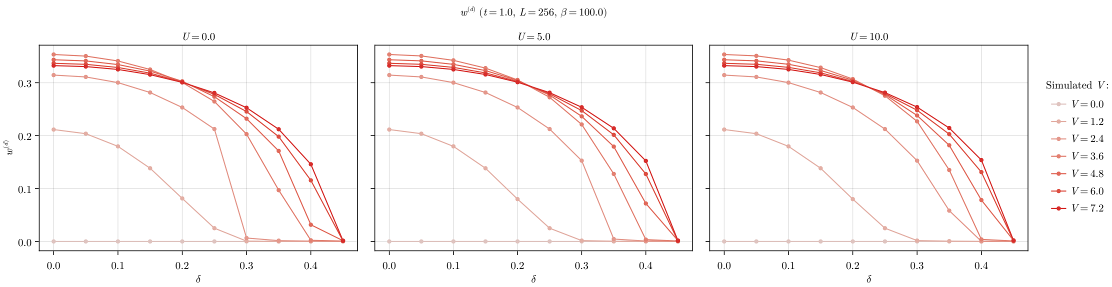
In Fig. 2.2b we see that the dependence on doping of $w^{(s^*)}$ is peculiar. At low doping ($\delta \lesssim 0.3$), $w^{(s^*)}$ is small for small values of V (left side of the three panels). At higher doping (right side) we see that a local maximum emerges, indicating that $w^{(s^*)}$ converges to higher values at smaller Vs.

d-wave gap. In Fig. 2.3 we report our findings for the HFP $w^{(d)}$, recalling that the d-wave gap is given by:

$$\tilde{\Delta}^{(d)} = V w^{(d)}$$



a | Dependency of $w^{(d)}$ on V at various Us, compared at three values of δ.



b | Dependency of $w^{(d)}$ on δ at various Vs, compared at three values of U.

Figure 2.3 | Dependency of the d-wave HFP $w^{(d)}$ on V at fixed U and δ (top panel, compared figures) and on δ at fixed V and U (bottom panel, compared figures).

We see from the top panel (Fig. 2.3a) that $w^{(d)}$ depends weakly on U. At half-filling (left panel), a d-wave gap is present at any $V > 0$:

$$\forall\ U/t \quad \wedge \quad V = 0 \quad : \quad \tilde{\Delta}^{(s)}\big|_{\delta=0} = 0$$

This is a key difference with respect to $w^{(s)}$, $w^{(s^*)}$. The self-consistent superconducting order parameter can be purely d-wave, or a mixture of d and s^* symmetries. In the first case, the gap changes sign when subject to rotations of angle $\pi/2$; in the second no symmetry is present under said rotations. At finite filling (central and right panels) a critical value $V_c^{(d)}$ emerges such that the d-wave gap appears at $V > V_c^{(d)}$.

The dependence of $w^{(d)}$ on doping at different values of U is shown in the bottom row (Fig. 2.3b). For any V, the HFP $w^{(d)}$ is a monotonically decreasing function of δ. As we said, the critical value $V_c^{(d)}$ increases with doping. Because of this, we see that the lighter curves of Fig. 2.3b, corresponding to lower values of V, reach zero quite rapidly. The curves for larger values of V, instead, reach zero at almost unitary filling. In all three panels of Fig. 2.3b we see that at doping $\delta \approx 0.3$ all lines intersect. In particular, decreasing V the curve gets steeper. At low doping, there exists a finite value of V that maximises $w^{(d)}$.

Competition among the channels. To get a valid understanding of the competition among channels, we need to compare the values of the harmonics $\tilde{\Delta}^{(\gamma)}$ for $\gamma = s, s^*, d$. Competition is reflected in the relative ratio between different components. We report a comparison of such in Fig. 2.4. **The presented data are limited to the case $U/t = 4$ for graphic reasons,but data and plots for other values of U are openly accessible at our repository.**

On the left column we plot three heatmaps, one for each harmonic in singlet SC phase. The same data are shown, superimposed, in the surface plot on the right. The harmonics $\tilde{\Delta}^{(\gamma)}$ cross if V/t gets large enough, with the d-wave component reaching the largest values in the parametric region we investigated, followed closely by the s^*-wave harmonic. Isotropic harmonics (s and s^*) are significantly non-zero in the same region, as we see in the first two rows of the left column. The two channel differ in amplitude significantly: the maximum value reached by the local gap

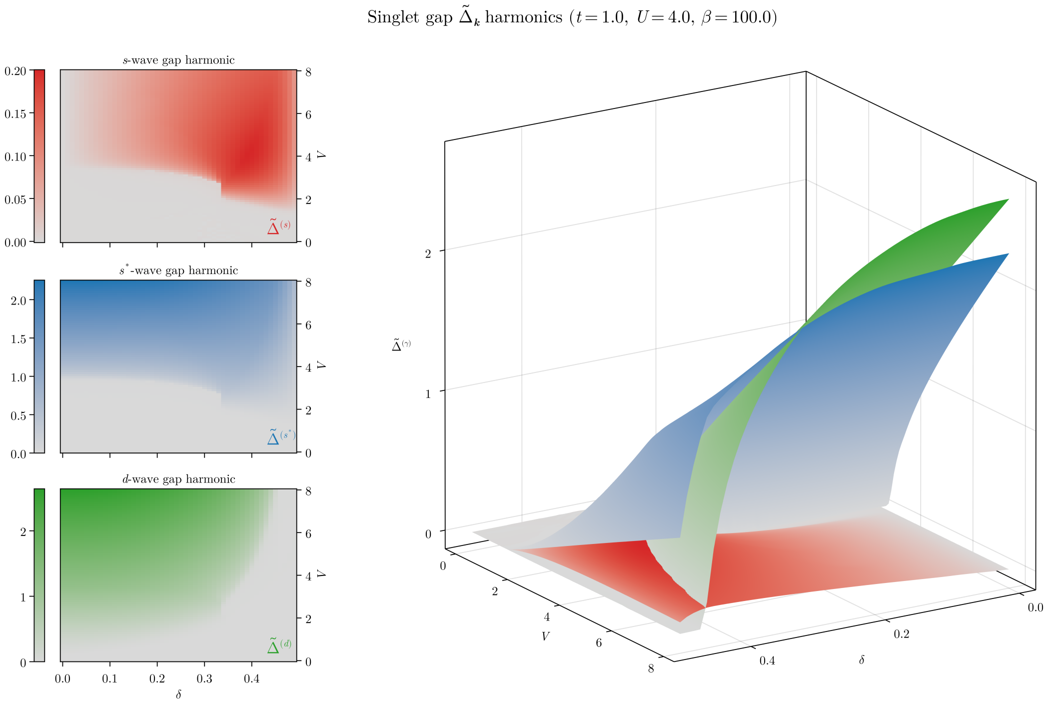


Figure 2.4 Plot of the singlet gap harmonics $\tilde{\Delta}^{(\gamma)}$, with $\gamma = s, s^*, d$ at $U/t = 4$. The panels on the left show the three harmonics separately as heatmaps, while the surface plot on the right shows them comparatively.

harmonic $\tilde{\Delta}^{(s)}$ is approximately an order of magnitude smaller than the maximum of the non-local harmonic $\tilde{\Delta}^{(s^*)}$. Indeed, the colorbars on the left of each heatmap is rendered such that the highest point corresponds to the maximum of the dataset. Moreover, we see that at $\delta = 0.0$ the local isotropic harmonic $\tilde{\Delta}^{(s)}$ is completely suppressed, as we expressed in Eq. (2.61).

All harmonics share a quite strange boundary around $\delta \approx 0.34$, $V/t \approx 2.5$, visible as a sharp vertical segment in the three heatmaps on the left column of Fig. 2.4. The isotropic harmonics suddenly change from zero to large values moving from left to right, while the d-wave part does the opposite. Whether this behaviour underlies some deeper structure we were not able to tackle with our coarse-grained simulations, or is a numeric artifact, we do not know.

Four phases can be distinguished: the normal state and three superconducting regions (pure d-wave, pure $s \oplus s^*$-wave, mixed). In Fig. 2.5 we give a schematic depiction of how the plane (δ, V) divides in the four regions at $U/t = 0$ (left), $U/t = 4$ (centre) and $U/t = 12$ (right). The boundaries were computed by setting an arbitrary threshold $\varepsilon = 10^{-2}$, with the following rule:

$$
\begin{array}{rcccc}
\text{Normal} & \left|\tilde{\Delta}^{(s)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(s^*)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(d)}\right| < \varepsilon \\
d\text{-wave} & \left|\tilde{\Delta}^{(s)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(s^*)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(d)}\right| \geq \varepsilon \\
s \oplus s^*\text{-wave} & \left[\left|\tilde{\Delta}^{(s)}\right| \geq \varepsilon \right. & \vee & \left. \left|\tilde{\Delta}^{(s^*)}\right| \geq \varepsilon\right] & \wedge & \left|\tilde{\Delta}^{(d)}\right| < \varepsilon \\
\text{Mixed} & \left[\left|\tilde{\Delta}^{(s)}\right| \geq \varepsilon \right. & \vee & \left. \left|\tilde{\Delta}^{(s^*)}\right| \geq \varepsilon\right] & \wedge & \left|\tilde{\Delta}^{(d)}\right| \geq \varepsilon
\end{array}
$$

We consider s and s^* as part of the same isotropic channel: just one harmonic being larger than the threshold is sufficient for us to assert that the gap function has an isotropic harmonic.

Comparing the three panels of Fig. 2.5, we see that increasing the value of the local repulsion U the shapes of the four regions change. In particular, the d-wave region expands to the right (larger δs); the normal-$s \oplus s^*$ SC boundary moves up (larger Vs). Note that on the left panel the isotropic region is labeled as purely s^*-wave: this is because if $U = 0$, the s-wave gap harmonic is null, even if $w^{(s)}$ converges to a non-zero value. The growth of the local repulsion U has two effects:

1. turning on the s-wave gap harmonic;

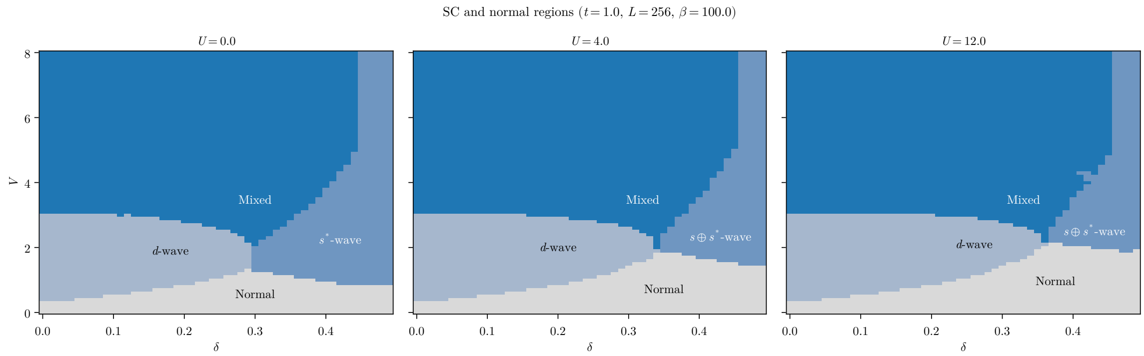


Figure 2.5 Depiction of the normal state region and the three singlet superconducting regions, at $U/t = 0, 4, 12$ from left to right. Lighter blue indicates the pure d-wave SC phase, blue the isotropic $s \oplus s^*$ phase, deeper blue the mixed region. In text is described the convention we used to identify the boundaries.

2. expanding the d-wave region and restricting the isotropic and mixed region.

From the last point we understand that increasing U we allow for the gap function to have a finite s-wave component, but this acts as a sort of internal competition within the isotropic region. Indeed, from the constraint of Eq. (2.60), we know that $w^{(s)} \neq 0$ only if $w^{(\gamma)} \neq 0$, with $\gamma \neq s$. In the $s \oplus s^*$ region the constraint reads:

$$\left|w^{(s^*)}\right|^2 \geq \left|w^{(s)}\right|^2 \frac{U}{V} \qquad \text{(Isotropic region)}$$

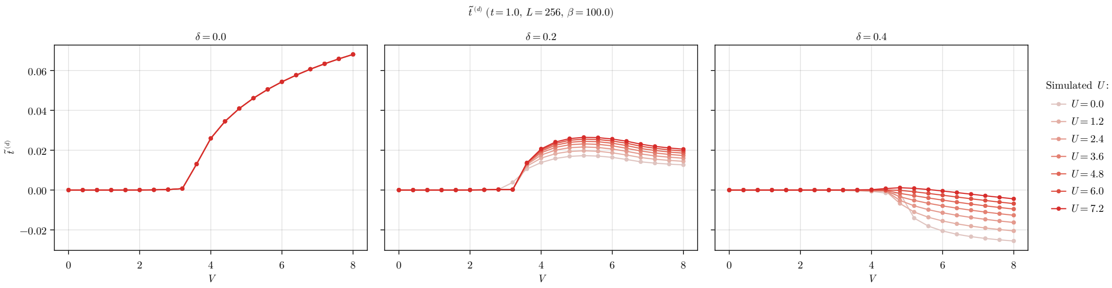
Suppose we work at fixed δ and $V > 0$ in the isotropic region at $U = 0$. There is no constraint whatsoever on $w^{(s^*)}$. Also, $w^{(s)}$ is free to converge at some finite value (we saw this in the left panel of Fig. 2.1b): the gap still has no s-wave component. Suppose now we switch on $U > 0$. Now, in order for $w^{(s)}$ to reach the same value as before, $w^{(s^*)}$ has to respect the constraint posed by the above equation. As U increases, the right-hand side gets bigger and $w^{(s^*)}$ needs to converge to larger and larger values to sustain $w^{(s)}$, otherwise to drop to zero. This is why increasing U the isotropic boundary moves up: $w^{(s^*)}$ isn't "free" anymore and must respect the constraint of Eq. (2.60).

In the three panels of Fig. 2.5, the mixing region is pretty much unchanged. At low doping, it is bounded from below by an horizontal line at $V/t \approx 3$. Increasing doping at fixed U, we see that the lower boundary reaches a cusp-like minimum around $\delta \approx 0.3 \div 0.4$ and then grows. This cusp coincides with the aforementioned sharp vertical segment, visible in Fig. 2.4. If we work at fixed doping and U, and choose δ such that we are aligned with the horizontal position of the cusp on the (δ, V) plane, the critical value of V/t for transitioning to a mixed SC state is minimised. At high values of V/t the mixing region extends to almost unitary doping, being limited on the right by a vertical line at $\delta \gtrsim 0.4$. We remind that we are discerning the SC phases by applying an arbitrary threshold ε to each gap component: we imagine that by using a more refined technique and by increasing the resolution of the simulation the boundary would appear less straight. Anyway, apart from a vertical stripe at almost unitary filling, at high V/t the dominant singlet SC phase is described by a mix of d-wave and $s \oplus s^*$-wave component. Recalling the surface plot of Fig. 2.4, we see that at $V/t \gg 1$ the s-wave component gets weaker with respect to the other two. In this limit, if $U \ll V$, the s-wave component is negligible, giving an approximate $d \oplus s^*$ mixed phase.

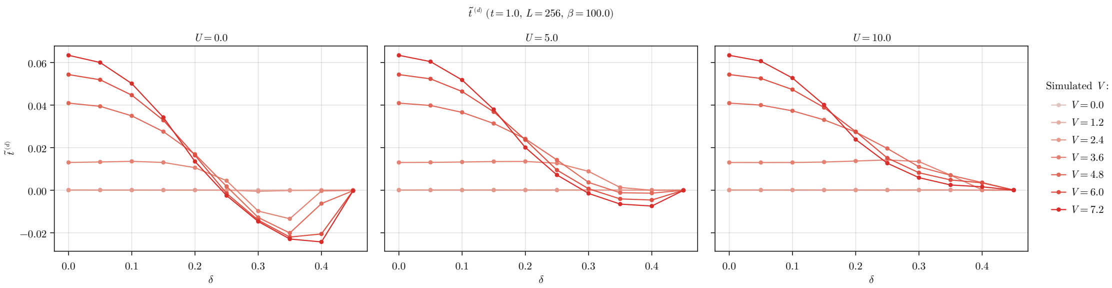
The mixing region is of particular interest also due its connection with the planar rotational SSB,

$$\text{Symmetry mixing} \quad \Longrightarrow \quad \mathrm{C}_4 \overset{\text{SSB}}{\to} \mathrm{C}_2$$

As we show in next section, the mixed-symmetry region coincides with the portion of the (δ, V) plane where planar rotational symmetry is broken.



a | Dependency of $\tilde{t}^{(d)}$ on V at various Us, compared at three values of δ.



b | Dependency of $\tilde{t}^{(d)}$ on δ at various Vs, compared at three values of U.

Figure 2.6 | Dependency of the d-wave renormalised hopping $t^{(d)}$ varying the doping δ and the local repulsion U. In the top panel we compare three doping levels $\delta = 0.0,\ 0.2,\ 0.4$ varying $0 \leq U \leq 8$. In the bottom panel we compare three repulsion values $U = 0.0,\ 5.0,\ 10.0$ varying $0 \leq \delta \leq 0.45$.

2.10.2 Rotational SSB and nematic superconductivity

In the context of this thesis, nematicity is established whenever $\tilde{t}^{(d)}$ becomes non-zero. This was discussed in detail in the context of the normal phase. If the renormalised hopping parameter has a non-zero d-wave component, this means that bare hopping is effectively boosted in one direction and damped in the other. Indeed, from Eq. (1.22), we know that

$$\tilde{t}^{(d)} = \frac{\tilde{t}^{(y)} - \tilde{t}^{(x)}}{2}$$

As we anticipated at the end of Sec. 1.3.3, if $\tilde{t}^{(d)} > 0$ the renormalised hopping is larger in direction y, if $\tilde{t}^{(d)} < 0$ instead it's larger in direction x. This interpretation serves to grasp the general idea behind nematicity, but is to be taken carefully. Let us get more rigorous. In the SC phase, the many body state is a gas of Bogoliubov quasiparticles which do not "hop on neighbouring sites". If $\tilde{t}^{(d)}$ is non-zero, the superconductor is nematic in the sense that Bogoliubov quasibands are not rotationally invariant,

$$\tilde{E}_{\mathbf{k}} \neq \tilde{E}_{\mathcal{R}\mathbf{k}}$$

being $\mathcal{R}$ a $\pi/2$ rotation. We connect this effect to hopping renormalisation *a posteriori* following the formal discussion of Sec. 1.3.3.

For the normal and AF phases we were able to prove that $u^{(d)} = 0$ for any choice of model parameters. For the SC phase, instead, $u^{(d)}$ converges to non-zero values. In Fig. 2.6, we plot the convergence value of

$$\tilde{t}^{(d)} = -\frac{V u^{(d)}}{2}$$

instead of the HFP $u^{(d)}$: nematicity is undoubtedly present. In the top panel (Fig. 2.6a), we see that a critical value of V exists over which nematicity is established. Moreover, the $\delta = 0.0$ case (left panel) is independent of U. In all cases, the convergence values for $\tilde{t}^{(d)}$ are two order of magnitudes smaller than $t = 1.0$: in the region of parametric space we explore, nematicity is a

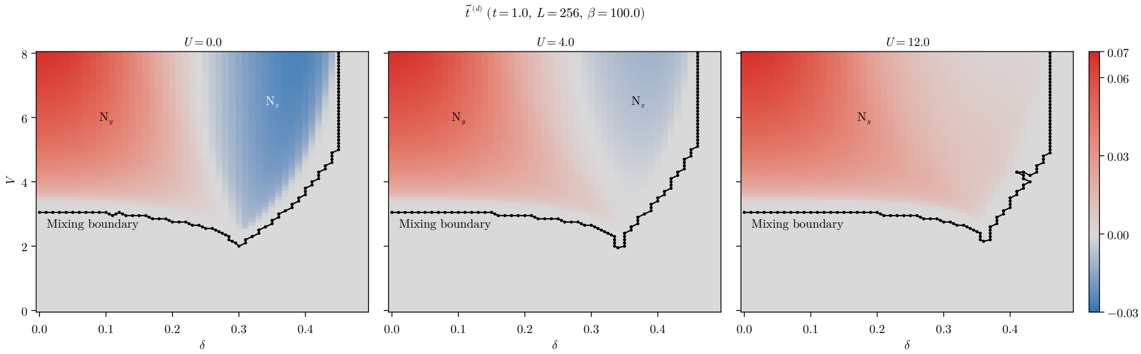


Figure 2.7 Heatmap of the anisotropic component of renormalised hopping, $\tilde{t}^{(d)}$, at $U/t = 0, 4, 12$ from left to right. The dataset is the same of Fig. 2.5, and the mixing boundary is the set of points separating the mixed region from the rest of the phases.

sharp but small effect. Note that at $\delta = 0.2$ (central panel) $\tilde{t}^{(d)} \geq 0$, while at $\delta = 0.4$ (right panel) $\tilde{t}^{(d)} \leq 0$. Increasing the doping we change the sign of $\tilde{t}^{(d)}$

This sign change is most evident in the first two figures of the bottom panel (Fig. 2.6b), where we vary the doping level while keeping fixed the local repulsion U. The behaviour of $\tilde{t}^{(d)}$ is similar at low doping in the three cases, but changes significantly at larger doping. Indeed, at $U/t = 0$ (left figure) and $U/t = 5$ (central figure) at some point $\tilde{t}^{(d)}$ changes sign. At $U/t = 10$ (right figure), all our data points are positive (possibly the sign flip occurs at higher Vs). Moreover, as U increases, the amplitude of the oscillation of $\tilde{t}^{(d)}$ as a function of δ reduces. We conclude that both the local repulsion and electronic doping disrupt nematicity, which is instead most effective for a purely NNs attractive model.

Nematic effects are directly related to symmetry mixing in the superconducting order parameter. Consider Fig. 2.7, where we plot the convergence values for $\tilde{t}^{(d)}$ in the (δ, V) plane for $U/t = 0, 4, 12$. The used dataset is the same of Fig. 2.5. The superimposed black lines are the mixing boundaries, the set of extremal points of the deeper blue regions of Fig. 2.5. Despite a bit of roughness related to the gridded nature of our data, it is evident that these boundaries encircle the regions where $\tilde{t}^{(d)} \neq 0$. Note that the set of points composing the mixing regions are those satisfying the threshold rule exposed in last section; choosing another threshold, the black curves change. Nevertheless, the key point is that there is an evident connection between symmetry mixing and breaking of planar rotational symmetry.

The mixing region coincides with the nematic region. As we said at the beginning of this section, nematicity can be established in two ways, based on the direction along which hopping is boosted. Hence it must hold

$$\mathrm{M} \simeq \mathrm{N}_x \cup \mathrm{N}_y$$

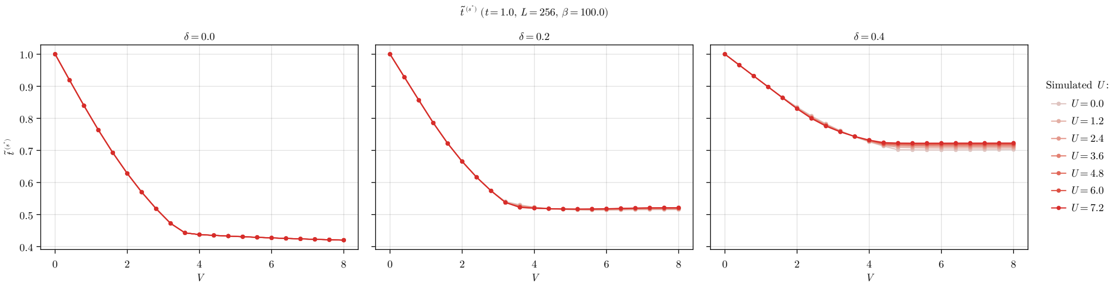
with the following convention on notation:

$$\begin{aligned}
\mathrm{M} &= \text{Region of } (\delta, V) \text{ plane where mixing occurs} \\
\mathrm{N}_x &= \text{Region of } (\delta, V) \text{ plane where } \tilde{t}^{(d)} < 0 \\
\mathrm{N}_y &= \text{Region of } (\delta, V) \text{ plane where } \tilde{t}^{(d)} > 0
\end{aligned}$$

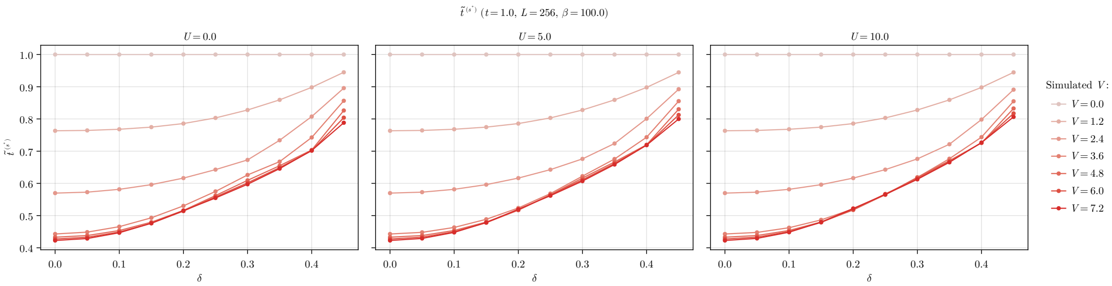
The notation N_ℓ has the following meaning: the index ℓ indicates the direction along which renormalised hopping is larger. We indicate in Fig. 2.7 these regions for $U/t = 0, 4, 12$.

Note that the map were colour-corrected in order to enhance the visibility of the blue region. The intensity of the colouring there suggests that the amplitude is comparable with the red region, but is not. Indeed, we set up the colormap such that gray is aligned with $\tilde{t}^{(d)} = 0.0$, and intensity of colour grows asimmetrically in the negative and positive direction.

The three panels of Fig. 2.7 differ crucially in the fact that the data for $U/t = 12$ (right) are all positive, while those for $U/t = 0, 4$ (left,centre) are negative in a portion of space. In other words, as we increase U/t, N_y expands and N_x shrinks. Moving from $U/t = 0$ (left) to $U/t = 4$ (centre) we see that the amplitude of $\tilde{t}^{(d)}$ decreases in N_x and not in N_y. We already observed this



a | Dependency of $\tilde{t}^{(s^*)}$ on V at various Us, compared at three values of δ.



b | Dependency of $\tilde{t}^{(s^*)}$ on δ at various Vs, compared at three values of U.

Figure 2.8 | Dependency of the s^*-wave renormalised hopping $t^{(s^*)}$ varying the doping δ and the local repulsion U. In the top panel we compare three doping levels $\delta = 0.0,\ 0.2,\ 0.4$ varying $0 \leq U \leq 8$. In the bottom panel we compare three repulsion values $U = 0.0,\ 5.0,\ 10.0$ varying $0 \leq \delta \leq 0.45$.

in Fig. 2.6b. Moreover, for $U/t = 12$ (right) N_x is apparently absent and N_y takes up the entirety of the mixing region. Whether there exists some critical value of U/t above which N_x disappears in the entirety of the (δ, V) plane, we do not know.

Breaking of planar rotational symmetry and the insurgence of a d-wave harmonic for the renormalised hopping is certainly the most peculiar effect for the singlet SC phase. It adds to the isotropic bands shrinking, which is, the s^*-wave renormalisation of the hopping parameter. We discuss it in next section.

2.10.3 Isotropic bands shrinking

The last HFP to be checked is $u^{(s^*)}$, responsible of bands shrinking. Its physical role is the isotropic renormalisation of the hopping parameter

$$\tilde{t}^{(s^*)} = t - \frac{V u^{(s^*)}}{2}$$

For the sake of conciseness, we skip plotting $u^{(s^*)}$ and plot directly $\tilde{t}^{(s^*)}$ in Fig. 2.8.

In the top panel (Fig. 2.8a), we show the dependence of $t^{(s^*)}$ on V. The three curves, obtained at three doping levels, are approximately independent of U. The common feature is that $\tilde{t}^{(s^*)}$ decreases somewhat linearly with V up to some critical value, above which it flattens to an horizontal asymptote. Electronic doping changes primarily the height of this asymptote

As was for the AF and normal phases, electronic doping disrupts the isotropic bands shrinking effect. At half-filling (left figure) we see that bands shrinking is most effective, reducing the hopping parameter to less than half its $V = 0$ value. In the bottom panel (Fig. 2.8b) we show $t^{(s^*)}$ as a function of δ, varying V and U. As we already noted, the dependence on U is very weak and the three figures are a lot alike. As we already noted, $t^{(s^*)}$ increases with doping.

In Fig. 2.9 we compare $t^{(s^*)}$ for the normal phase (blue surface) and the SC phase (green surface). The two surfaces share a general feature: $t^{(s^*)}$ is approximately unitary at low V/t and almost unitary doping, and decreases significantly in the low-δ and high-V/t limit. In this

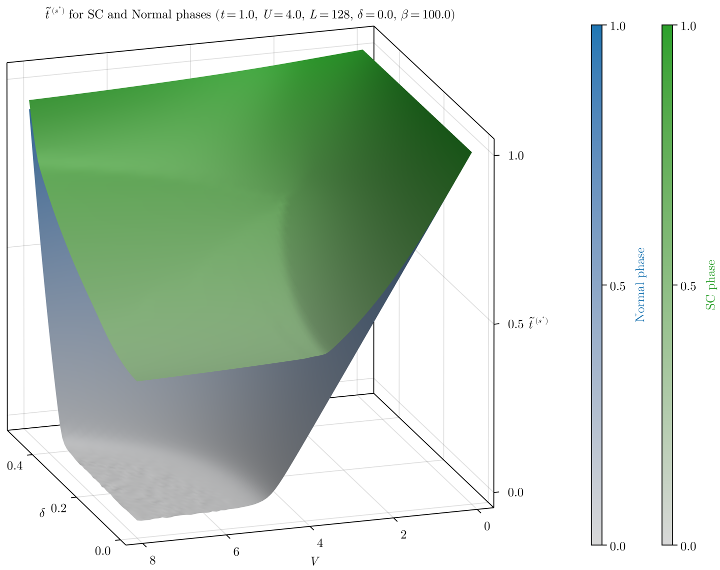


Figure 2.9 Comparison of the isotropic part of the renormalised hopping $\tilde{t}^{(s^*)}$ for the SC (green) and normal (blue) phases on the (δ, V) plane. The data from the normal phase are the same of Fig. 1.7.

region, for the normal phase we observe the Mott-localisation plateau discussed in Sec. 1.6.1, where $t^{(s^*)} \approx 0$. In the SC phase, looking at the same region, $t^{(s^*)}$ is minimised but no plateau is formed. More importantly, isotropic hopping is reduced to roughly half the value of t and not nullified. We do not observe a Mott-like area as for the normal phase.

Recalling the plots of Fig. 2.5, the low-δ and high-V/t limit (the corner where $t^{(s^*)}$ is minimised) falls deep within the mixed region. Recalling also Fig. 2.4, in this corner the d-wave and s^*-wave harmonics of the gap function are large and comparable. Finally, we know from Fig. 2.7 (central panel) that here nematic effects are small, but significantly non-zero. In particular, considering that

$$\lim_{\delta\to 0}\lim_{V/t\gg 1}\frac{t^{(s^*)}}{t} \approx 0.4 \qquad \lim_{\delta\to 0}\lim_{V/t\gg 1}\frac{t^{(d)}}{t} \approx 0.06$$

we see that the two components differ just by one order of magnitude. Nematic effects account for a rather large portion of hopping renormalisation.

With this consideration, we end the presentation of HFPs and RMPs and move to the last section, where we discuss the results regarding free energy and entropy densities.

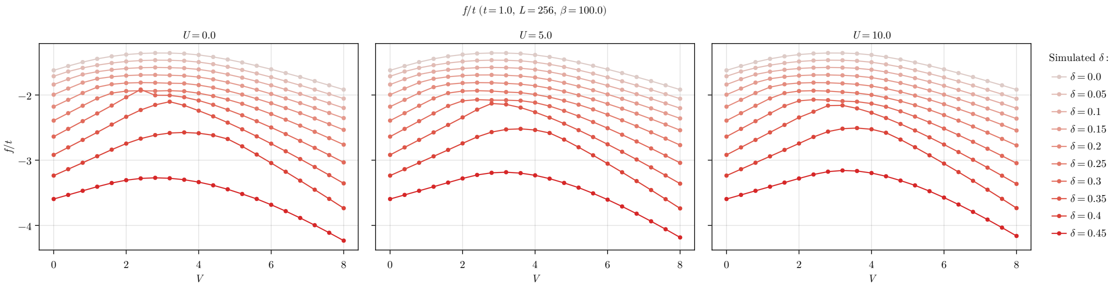
2.10.4 Solution free energy and entropy densities

For the SC phase, free energy density is given by Eq. (2.58). Entropy density is obtained by Eq. (2.57) using that $s = 2s^{(+)}$, as we argued in the sequent lines.

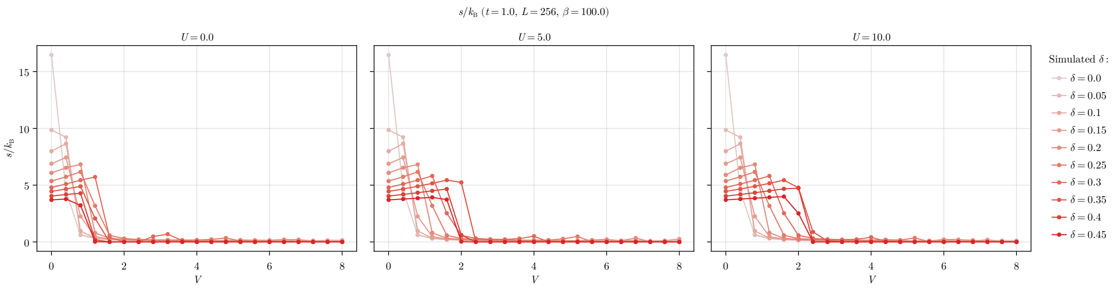
In Fig. 2.10 we show the data for f and s varying

2.11 A final note

[To be continued...]



a | Dependency of f on V at various Us, compared at three values of δ.



b | Dependency of s on V at various δs, compared at three values of U.

Figure 2.10 | Dependency of free energy (top) and entropy (bottom) densities on V varying the doping δ and the local repulsion U.

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